

ANALYSIS AND NUMERICAL INVESTIGATION OF NONLINEAR
SEMICONDUCTOR DRIFT-DIFFUSION SYSTEMS

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ABSTRACT

Nonlinear drift–diffusion systems play a fundamental role in the mathematical modeling of semiconductor devices. This thesis presents an analytical and numerical investigation of one-dimensional semiconductor p–n junction devices governed by coupled Poisson–Boltzmann and drift–diffusion equations.

In Chapter 2, finite-difference discretizations and iterative solution methods are developed for the nonlinear Poisson–Boltzmann equation. Analytical properties of the resulting discrete systems are investigated through contraction mapping arguments and spectral analysis of the discrete Laplacian, leading to an interpretation of the convergence condition in terms of the Debye screening length.

In Chapter 3, the nonlinear solver is further studied through numerical simulation, and the quasi-linearized formulation is shown to be mathematically equivalent to Newton’s method applied to the discrete Poisson–Boltzmann system. The framework is subsequently extended through quasi-Fermi and Slotboom reformulations of the drift–diffusion equations and through the Scharfetter–Gummel discretization.

In Chapter 4, the framework is extended to non-equilibrium drift–diffusion systems under applied bias conditions. Using a quasi-linearized Gummel iteration together with Scharfetter–Gummel discretization, numerical simulations are performed to investigate forward- and reverse-bias operation, recombination and generation dynamics, current conservation, and the current–voltage characteristics of semiconductor p–n junction devices.

Finally, Chapter 5 discusses the possible extensions of the present framework and summarizes the main conclusion of this project.

DEDICATION

Dedicated to my future self, wherever mathematics may lead him, and to the curiosity that continues to make the journey worthwhile.

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LIST OF SYMBOLS

$\frac{\partial}{\partial t}$	Partial derivative with respect to time
$\frac{d}{dx}$	Derivative with respect to the spatial coordinate x
λ	Eigenvalue
μ_n	Electron mobility [$\text{cm}^2/(\text{V s})$]
μ_p	Hole mobility [$\text{cm}^2/(\text{V s})$]
∇	Gradient operator
$\nabla \cdot$	Divergence operator
∇^2	Laplacian operator
ϕ	Electrostatic potential [V]
ϕ_L	Left boundary potential [V]
ϕ_n	Electron quasi-Fermi potential [V]
ϕ_p	Hole quasi-Fermi potential [V]
ϕ_R	Right boundary potential [V]
ψ	Dimensionless potential jump
ρ	Net charge density [C/cm^3]
τ_n	Electron recombination lifetime [s]
τ_p	Hole recombination lifetime [s]
ε	Permittivity of the semiconductor material [F/cm]
ε_n	Electron concentration convergence tolerance
ε_p	Hole concentration convergence tolerance
ε_V	Electrostatic potential convergence tolerance
$B(\psi)$	Bernoulli function evaluated at the dimensionless potential jump
$C(x)$	Net doping profile [cm^{-3}]
D_n	Electron diffusion coefficient [cm^2/s]
D_p	Hole diffusion coefficient [cm^2/s]
E	Electric field [V/cm]
J_n	Electron current density [A/cm^2]
J_p	Hole current density [A/cm^2]
J_s	Reverse saturation current density [A/cm^2]
J_{Shockley}	Current density predicted by the Shockley diode law [A/cm^2]
$J_{n,i+\frac{1}{2}}^{\text{QF}}$	Discrete quasi-Fermi electron flux [A/cm^2]
$J_{n,i+\frac{1}{2}}^{\text{SG}}$	Scharfetter–Gummel electron flux [A/cm^2]
$J_{p,i+\frac{1}{2}}^{\text{QF}}$	Discrete quasi-Fermi hole flux [A/cm^2]
$J_{p,i+\frac{1}{2}}^{\text{SG}}$	Scharfetter–Gummel hole flux [A/cm^2]
k_B	Boltzmann constant [J/K]
L	Length of the computational domain [cm]
L_D	Debye screening length [cm]
L_F	Lipschitz constant
L_n	Electron diffusion length [cm]

L_p	Hole diffusion length [cm]
$L_{n,i+\frac{1}{2}}$	Logarithmic mean of the electron density [cm^{-3}]
$L_{p,i+\frac{1}{2}}$	Logarithmic mean of the hole density [cm^{-3}]
n	Electron concentration [cm^{-3}]
N_A^-	Ionized acceptor concentration [cm^{-3}]
N_D^+	Ionized donor concentration [cm^{-3}]
n_i	Intrinsic carrier concentration [cm^{-3}]
p	Hole concentration [cm^{-3}]
q	Elementary charge [C]
R	Net recombination/generation rate [$\text{cm}^{-3}\text{s}^{-1}$]
T	Absolute temperature [K]
t	Time [s]
$T[\phi]$	Fixed-point operator acting on the electrostatic potential
u	Electron Slotboom variable
v	Hole Slotboom variable
V_T	Thermal voltage [V]
V_{app}	Applied voltage [V]
V_{bi}	Built-in potential [V]
V_{eff}	Effective junction potential [V]
W_d	Depletion width [cm]
x	Spatial coordinate [cm]

1. Introduction

1.1. Semiconductor Background & Motivation. Semiconductor device modeling provides a rich interaction between nonlinear partial differential equations, numerical analysis, and physics. In particular, p–n junction devices serve as a fundamental setting in which electrostatic interactions, carrier transport, and recombination mechanisms become strongly coupled through nonlinear systems of equations. Understanding the mathematical structure and numerical behavior of such systems is therefore important in both applied mathematics and semiconductor physics [06]

1.2. Drift–Diffusion and Poisson–Boltzmann Models. One of the classical approaches used in semiconductor modeling is the drift–diffusion framework [VR50], where electron and hole transport are coupled with the electrostatic potential through Poisson’s equation. Under equilibrium assumptions, this coupling naturally leads to nonlinear Poisson–Boltzmann systems containing exponential carrier relations because of which, analytical solutions are generally intractable. And hence, numerical methods play a central role in studying the behavior of the system.

1.3. Objectives and Scope of the Project. The primary goal of this thesis is to study the mathematical modeling and numerical simulation of one-dimensional semiconductor p–n junction devices governed by coupled Poisson–Boltzmann and drift–diffusion equations. The work focuses on understanding how electrostatic interactions, carrier transport, recombination effects, and applied bias influence the behavior of semiconductor systems.

2. Mathematical Formulation & Analysis

2.1. Physical Background. Semiconductors are the materials whose electrical conductivity lies in between that of conductors and insulators. Their electric behavior can be influenced by external influences such as doping, applied bias, and so on. Because of these properties, semiconductors form the foundation of many modern electronic devices.

Charge transport in semiconductor is governed by two types of charged carriers: electrons & holes. Electrons are negatively charged carriers and are mostly in conduction band. Holes represents the absence of electrons in valence band and behave effectively as positively charged carriers. The movement and interaction of these charged carriers determine the electrical behavior of semiconductor devices [VR50].

2.2. Drift–Diffusion Formulation. When a p–type semiconductor is connected to an n–type semiconductor, holes from the p–type region tend to move toward the n–type region, while electrons from the n–type region move toward the p–type region. This motion, caused by differences in carrier concentration, is known as diffusion. As charge carriers move across the junction, a region of charge imbalance is formed near the interface between the two materials. This separation of charges creates a built-in electric potential, also known as the potential barrier.

The resulting electric field opposes further diffusion by pushing holes and pulling electrons in the opposite direction. This type of carrier motion caused by the electric field is known as drift. The combined effects of drift and diffusion give rise to drift–diffusion system. The primary unknowns are the electron density n , the hole density p , and the electrostatic potential ϕ .

The electron and hole current densities are given by[VR50, V GK17]

$$(2.1) \quad J_n = -q\mu_n n \nabla \phi + qD_n \nabla n,$$

$$(2.2) \quad J_p = -q\mu_p p \nabla \phi - qD_p \nabla p,$$

where q denotes the elementary charge, μ_n, μ_p are the electron and hole mobilities, and D_n, D_p are the corresponding diffusion coefficients.

The carrier densities satisfy the continuity equations

$$(2.3) \quad \frac{\partial n}{\partial t} = \frac{1}{q} \nabla \cdot J_n - R,$$

$$(2.4) \quad \frac{\partial p}{\partial t} = -\frac{1}{q} \nabla \cdot J_p - R,$$

where R denotes the net recombination rate.

Substituting the current densities into the continuity equations gives

$$(2.5) \quad \frac{\partial n}{\partial t} = \nabla \cdot (D_n \nabla n - \mu_n n \nabla \phi) - R,$$

$$(2.6) \quad \frac{\partial p}{\partial t} = \nabla \cdot (D_p \nabla p + \mu_p p \nabla \phi) - R.$$

The electrostatic potential satisfies Poisson's equation [06].,

$$(2.7) \quad -\nabla \cdot (\varepsilon \nabla \phi) = q (p - n + N_D^+ - N_A^-).$$

where ε denotes the permittivity of the material, while N_D^+ and N_A^- represent the ionized donor and acceptor concentrations.

If the permittivity is assumed constant, then Poisson's equation reduces to

$$(2.8) \quad \nabla^2 \phi = -\frac{q}{\varepsilon} (p - n + N_D^+ - N_A^-).$$

Thus, the general non-equilibrium drift-diffusion-Poisson system is given by

$$(2.9) \quad \frac{\partial n}{\partial t} = \nabla \cdot (D_n \nabla n - \mu_n n \nabla \phi) - R,$$

$$(2.10) \quad \frac{\partial p}{\partial t} = \nabla \cdot (D_p \nabla p + \mu_p p \nabla \phi) - R,$$

$$(2.11) \quad \nabla^2 \phi = -\frac{q}{\varepsilon} (p - n + N_D^+ - N_A^-).$$

Under thermodynamic equilibrium, the carrier densities do not vary with time, so

$$\frac{\partial n}{\partial t} = \frac{\partial p}{\partial t} = 0.$$

If recombination effects are neglected by assuming

$$R = 0,$$

then the system reduces to

$$(2.12) \quad \nabla \cdot (D_n \nabla n - \mu_n n \nabla \phi) = 0,$$

$$(2.13) \quad \nabla \cdot (D_p \nabla p + \mu_p p \nabla \phi) = 0,$$

$$(2.14) \quad \nabla^2 \phi = -\frac{q}{\varepsilon} (p - n + N_D^+ - N_A^-).$$

This is the steady-state drift-diffusion-Poisson model.

2.3. The Poisson-Boltzmann Equation.

We now reduce the drift-diffusion-Poisson system to one spatial dimension. Assume that all unknowns depend only on the spatial variable x . Then

$$n = n(x), \quad p = p(x), \quad \phi = \phi(x).$$

The steady-state carrier equations become

$$(2.15) \quad \frac{d}{dx} \left(D_n \frac{dn}{dx} - \mu_n n \frac{d\phi}{dx} \right) = 0,$$

$$(2.16) \quad \frac{d}{dx} \left(D_p \frac{dp}{dx} + \mu_p p \frac{d\phi}{dx} \right) = 0.$$

Integrating once gives

$$(2.17) \quad D_n \frac{dn}{dx} - \mu_n n \frac{d\phi}{dx} = C_n,$$

$$(2.18) \quad D_p \frac{dp}{dx} + \mu_p p \frac{d\phi}{dx} = C_p.$$

At thermodynamic equilibrium, there is no net carrier current through the device.

Therefore,

$$C_n = 0, \quad C_p = 0.$$

Thus,

$$(2.19) \quad D_n \frac{dn}{dx} = \mu_n n \frac{d\phi}{dx},$$

$$(2.20) \quad D_p \frac{dp}{dx} = -\mu_p p \frac{d\phi}{dx}.$$

Using the Einstein relation [06],

$$D_n = \mu_n V_T, \quad D_p = \mu_p V_T,$$

where

$$V_T = \frac{k_B T}{q}$$

is the thermal voltage, k_B is Boltzmann's constant, T is absolute temperature, and q is the elementary charge, we obtain

$$(2.21) \quad V_T \frac{dn}{dx} = n \frac{d\phi}{dx},$$

$$(2.22) \quad V_T \frac{dp}{dx} = -p \frac{d\phi}{dx}.$$

Dividing by n and p , respectively, gives

$$(2.23) \quad \frac{1}{n} \frac{dn}{dx} = \frac{1}{V_T} \frac{d\phi}{dx},$$

$$(2.24) \quad \frac{1}{p} \frac{dp}{dx} = -\frac{1}{V_T} \frac{d\phi}{dx}.$$

Equivalently,

$$(2.25) \quad \frac{d}{dx} (\ln n) = \frac{1}{V_T} \frac{d\phi}{dx},$$

$$(2.26) \quad \frac{d}{dx} (\ln p) = -\frac{1}{V_T} \frac{d\phi}{dx}.$$

Integrating again, we obtain

$$(2.27) \quad \ln n(x) = \frac{\phi(x)}{V_T} + C_1,$$

$$(2.28) \quad \ln p(x) = -\frac{\phi(x)}{V_T} + C_2.$$

Exponentiating both equations gives

$$(2.29) \quad n(x) = n_* e^{\frac{\phi(x)}{V_T}},$$

$$(2.30) \quad p(x) = p_* e^{-\frac{\phi(x)}{V_T}}.$$

where $n_* = e^{C_1}$ and $p_* = e^{C_2}$ are positive constants determined by reference conditions or boundary data.

The one-dimensional Poisson equation is

$$\frac{d^2 \phi}{dx^2} = -\frac{q}{\varepsilon} (p - n + N_D^+ - N_A^-).$$

Substituting the Boltzmann relations into Poisson's equation yields

$$(2.31) \quad \frac{d^2 \phi}{dx^2} = -\frac{q}{\varepsilon} \left[p_* e^{-\frac{\phi}{V_T}} - n_* e^{\frac{\phi}{V_T}} + N_D^+ - N_A^- \right].$$

Equation (2.31) is the one-dimensional Poisson–Boltzmann equation. Its nonlinearity comes from the exponential dependence of the carrier densities on the electrostatic potential.

2.4. Finite Difference Discretization. We consider the one-dimensional Poisson–Boltzmann equation

$$\frac{d^2\phi}{dx^2} = -\frac{q}{\varepsilon} \left[p_* e^{-\frac{\phi}{V_T}} - n_* e^{\frac{\phi}{V_T}} + N_D(x) - N_A(x) \right].$$

Let the computational domain be $[0, L]$, where L denotes the length of the semiconductor device. We introduce a uniform grid

$$x_i = ih, \quad h = \frac{L}{m}, \quad i = 0, 1, \dots, m.$$

Let

$$\phi_i \approx \phi(x_i).$$

For an interior node x_i , where $i = 1, \dots, m-1$, the second derivative is approximated by the centered finite difference formula [BFB15]

$$\frac{d^2\phi}{dx^2}(x_i) = \frac{\phi_{i+1} - 2\phi_i + \phi_{i-1}}{h^2} + O(h^2).$$

Ignoring the truncation error term gives the discrete approximation

$$\frac{\phi_{i+1} - 2\phi_i + \phi_{i-1}}{h^2} = -\frac{q}{\varepsilon} \left[p_* e^{-\frac{\phi_i}{V_T}} - n_* e^{\frac{\phi_i}{V_T}} + N_D(x_i) - N_A(x_i) \right],$$

for $i = 1, \dots, m-1$.

Equivalently, the interior finite difference equations can be written as

$$\frac{\phi_{i+1} - 2\phi_i + \phi_{i-1}}{h^2} = -\frac{q}{\varepsilon} \left[p_* e^{-\frac{\phi_i}{V_T}} - n_* e^{\frac{\phi_i}{V_T}} + N_D(x_i) - N_A(x_i) \right]$$

This produces a nonlinear algebraic equation at each interior grid point. The boundary conditions and the resulting matrix formulation will be introduced in the next section.

2.5. Hyperbolic Formulation and Boundary Conditions. Using the Boltzmann relations derived earlier,

$$n = n_* e^{\frac{\phi}{V_T}}, \quad p = p_* e^{-\frac{\phi}{V_T}},$$

we obtain

$$np = n_* p_*.$$

At thermal equilibrium,

$$np = n_i^2,$$

where n_i denotes the intrinsic carrier concentration. Therefore,

$$n_* p_* = n_i^2.$$

In the equilibrium setting considered here, we take

$$n_* = p_* = n_i.$$

Hence,

$$n = n_i e^{\frac{\phi}{V_T}}, \quad p = n_i e^{-\frac{\phi}{V_T}}.$$

Therefore,

$$p - n = n_i e^{-\frac{\phi}{V_T}} - n_i e^{\frac{\phi}{V_T}} = -2n_i \sinh\left(\frac{\phi}{V_T}\right).$$

Since

$$C(x) = N_D(x) - N_A(x),$$

the one-dimensional Poisson–Boltzmann equation

$$\frac{d^2 \phi}{dx^2} = -\frac{q}{\varepsilon} (p - n + C(x))$$

can be rewritten as

$$\frac{d^2 \phi}{dx^2} = \frac{2qn_i}{\varepsilon} \sinh\left(\frac{\phi}{V_T}\right) - \frac{q}{\varepsilon} C(x).$$

Defining

$$\lambda = \frac{2qn_i}{\varepsilon},$$

we obtain the compact hyperbolic formulation

$$\frac{d^2 \phi}{dx^2} = \lambda \sinh\left(\frac{\phi}{V_T}\right) - \frac{q}{\varepsilon} C(x).$$

Now, we impose Dirichlet boundary conditions

$$\phi(0) = \phi_L, \quad \phi(L) = \phi_R.$$

The Dirichlet structure specifies the value of the electrostatic potential at the two contacts, while the actual boundary values are determined from the semiconductor

physics at the contacts.

$$C(x) = N_D(x) - N_A(x).$$

In a p-type region, $C(x) < 0$, so holes dominate and

$$p \approx |C(x)|, \quad n \approx 0.$$

Using

$$p = n_i e^{-\frac{\phi}{V_T}},$$

we obtain

$$\phi \approx -V_T \ln \left(\frac{|C(x)|}{n_i} \right).$$

Thus, if the left contact is p-type, then

$$\phi_L = \phi(0) \approx -V_T \ln \left(\frac{|C(0)|}{n_i} \right).$$

In an n-type region, $C(x) > 0$, so electrons dominate and

$$n \approx C(x), \quad p \approx 0.$$

Using

$$n = n_i e^{\frac{\phi}{V_T}},$$

we obtain

$$\phi \approx V_T \ln \left(\frac{C(x)}{n_i} \right).$$

Thus, if the right contact is n-type, then

$$\phi_R = \phi(L) \approx V_T \ln \left(\frac{C(L)}{n_i} \right).$$

Therefore, for a p–n device with p-type material on the left and n-type material on the right, the Ohmic Dirichlet boundary conditions [VGK17] are

$$\phi(0) \approx -V_T \ln \left(\frac{|C(0)|}{n_i} \right), \quad \phi(L) \approx V_T \ln \left(\frac{C(L)}{n_i} \right).$$

2.6. Matrix Formulation & Fixed-Point Mapping. From the hyperbolic formulation,

$$\frac{d^2 \phi}{dx^2} = \lambda \sinh \left(\frac{\phi}{V_T} \right) - \frac{q}{\varepsilon} C(x),$$

the finite difference approximation at an interior node x_i gives

$$\frac{\phi_{i+1} - 2\phi_i + \phi_{i-1}}{h^2} = \lambda \sinh\left(\frac{\phi_i}{V_T}\right) - \frac{q}{\varepsilon} C(x_i),$$

for $i = 1, \dots, m-1$.

Let

$$\Phi = \begin{bmatrix} \phi_0 \\ \phi_1 \\ \vdots \\ \phi_{m-1} \\ \phi_m \end{bmatrix}.$$

Imposing the boundary conditions in the first and last rows, the discrete system can

be written as

$$\begin{bmatrix} 1 & 0 & 0 & \cdots & 0 & 0 \\ 1 & -2 & 1 & \cdots & 0 & 0 \\ 0 & 1 & -2 & \ddots & 0 & 0 \\ \vdots & \vdots & \ddots & \ddots & 1 & 0 \\ 0 & 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & \cdots & 0 & 1 \end{bmatrix} \begin{bmatrix} \phi_0 \\ \phi_1 \\ \vdots \\ \phi_{m-1} \\ \phi_m \end{bmatrix} = h^2 \begin{bmatrix} -V_T \ln\left(\frac{|C(0)|}{n_i}\right) \\ \lambda \sinh\left(\frac{\phi_1}{V_T}\right) - \frac{q}{\varepsilon} C(x_1) \\ \vdots \\ \lambda \sinh\left(\frac{\phi_{m-1}}{V_T}\right) - \frac{q}{\varepsilon} C(x_{m-1}) \\ V_T \ln\left(\frac{C(L)}{n_i}\right) \end{bmatrix}.$$

Equivalently, we write this nonlinear algebraic system in operator form as

$$A\Phi = F[\Phi],$$

where A is the finite difference matrix and $F[\Phi]$ contains the nonlinear hyperbolic sine term, the doping profile, and the boundary values.

Since the unknown vector consists of the grid values

$$\Phi = [\phi_0 \quad \phi_1 \quad \cdots \quad \phi_m]^T,$$

the finite difference matrix A defines a linear map from $\mathbb{R}^{m+1}$ to $\mathbb{R}^{m+1}$. With boundary conditions imposed in the first and last rows, the boundary values are fixed, while the interior rows correspond to the standard second-difference operator. If $A\Phi = 0$, then the boundary rows force $\phi_0 = 0$ and $\phi_m = 0$, and the interior equations force the grid values to be linear (second derivative = 0) between the endpoints. Since both endpoint values are zero, the only solution is $\Phi = 0$, giving a trivial kernel. Therefore, A is invertible.

Hence, the system can be rewritten as

$$\Phi = A^{-1}F[\Phi].$$

This motivates the fixed-point mapping

$$T[\Phi] = A^{-1}F[\Phi].$$

Therefore, solving the nonlinear discrete Poisson–Boltzmann equation is equivalent to finding a fixed point

$$\Phi = T[\Phi].$$

Starting from an initial guess $\Phi^{(0)}$, the fixed-point iteration is given by

$$\Phi^{(k+1)} = T[\Phi^{(k)}] = A^{-1}F[\Phi^{(k)}].$$

2.7. Contraction Mapping Analysis. We now analyze the fixed-point map associated with the discrete nonlinear Poisson–Boltzmann equation. Recall from the previous subsection that the nonlinear system can be written in the form

$$T[\Phi] = A^{-1}F[\Phi].$$

A solution of the nonlinear system corresponds to a fixed point

$$\Phi = T[\Phi].$$

Let $\Phi_1, \Phi_2 \in \mathbb{R}^{m-1}$. Then

$$\begin{aligned} T[\Phi_1] - T[\Phi_2] &= A^{-1}F[\Phi_1] - A^{-1}F[\Phi_2] \\ &= A^{-1}(F[\Phi_1] - F[\Phi_2]), \end{aligned}$$

since A^{-1} is linear. Taking the Euclidean norm gives

$$(2.32) \quad \|T[\Phi_1] - T[\Phi_2]\|_2 \leq \|A^{-1}\|_2 \|F[\Phi_1] - F[\Phi_2]\|_2.$$

It remains to estimate the nonlinear term. For the interior nodes,

$$F_i[\Phi] = \lambda \sinh\left(\frac{\phi_i}{V_T}\right), \quad i = 1, \dots, m-1.$$

Differentiating gives

$$\frac{\partial F_i}{\partial \phi_i} = \frac{\lambda}{V_T} \cosh\left(\frac{\phi_i}{V_T}\right).$$

Since the hyperbolic cosine is unbounded on $\mathbb{R}$, the nonlinear map F is not globally Lipschitz. Therefore, we restrict the analysis to the bounded set

$$S_M = \left\{ \Phi \in \mathbb{R}^{m-1} : \|\Phi\|_\infty \leq M \right\}.$$

For $\Phi \in S_M$, each component satisfies $|\phi_i| \leq M$. Hence

$$\left| \frac{\partial F_i}{\partial \phi_i} \right| \leq \frac{\lambda}{V_T} \cosh \left(\frac{M}{V_T} \right).$$

Define

$$L_F = \frac{\lambda}{V_T} \cosh \left(\frac{M}{V_T} \right).$$

Now let $\Phi_1, \Phi_2 \in S_M$. By the Mean Value Theorem, for each interior component,

$$|F_i[\Phi_1] - F_i[\Phi_2]| \leq L_F |\phi_{1,i} - \phi_{2,i}|.$$

Squaring both sides and summing over all interior nodes gives

$$\sum_{i=1}^{m-1} |F_i[\Phi_1] - F_i[\Phi_2]|^2 \leq L_F^2 \sum_{i=1}^{m-1} |\phi_{1,i} - \phi_{2,i}|^2.$$

Taking square roots, we obtain

$$(2.33) \quad \|F[\Phi_1] - F[\Phi_2]\|_2 \leq L_F \|\Phi_1 - \Phi_2\|_2.$$

Therefore, from (2.32) and (2.33), we get

$$\|T[\Phi_1] - T[\Phi_2]\|_2 \leq \|A^{-1}\|_2 L_F \|\Phi_1 - \Phi_2\|_2.$$

Thus, T is a contraction on S_M provided

$$\|A^{-1}\|_2 L_F < 1.$$

Equivalently,

$$\|A^{-1}\|_2 \frac{\lambda}{V_T} \cosh \left(\frac{M}{V_T} \right) < 1.$$

Under this sufficient condition, Banach's Fixed Point Theorem [Eva98] implies that T has a unique fixed point in S_M , and the fixed-point iteration converges to that fixed point.

This condition has a beautiful interpretation. The hyperbolic sine term represents the nonlinear forcing produced by the carrier densities, while the inverse discrete Laplacian acts as the smoothing part of the elliptic problem. The contraction condition requires the smoothing effect of the discrete operator to dominate the local growth of

the nonlinear term. This motivates a closer study of the spectral properties of the discrete Laplacian.

2.8. Spectral Analysis of the Discrete Laplacian. The contraction condition from the previous subsection shows that convergence depends on the balance between the inverse discrete Laplacian and the local growth of the nonlinear term. Since the Euclidean norm is used in the contraction estimate, the spectral properties of the discrete Laplacian can be analyzed directly through its eigenvalues.

Let the semiconductor device occupy the interval $[0, L]$, where L denotes the device length. Choose a uniform grid

$$x_i = ih, \quad h = \frac{L}{m}, \quad i = 0, 1, \dots, m.$$

For the spectral analysis, we focus on the interior grid values

$$\Phi = [\phi_1 \ \phi_2 \ \cdots \ \phi_{m-1}]^T \in \mathbb{R}^{m-1}.$$

The second-difference operator is

$$(A\Phi)_i = \frac{\phi_{i+1} - 2\phi_i + \phi_{i-1}}{h^2}, \quad i = 1, \dots, m-1.$$

Thus,

$$A = \frac{1}{h^2} \begin{bmatrix} -2 & 1 & 0 & \cdots & 0 \\ 1 & -2 & 1 & \ddots & \vdots \\ 0 & 1 & -2 & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & 1 \\ 0 & \cdots & 0 & 1 & -2 \end{bmatrix}.$$

The matrix A is symmetric and negative definite. A standard result for tridiagonal Toeplitz matrices gives the eigenvalues [LeV07]

$$\lambda_k(A) = -\frac{4}{h^2} \sin^2 \left(\frac{k\pi}{2m} \right), \quad k = 1, \dots, m-1.$$

Therefore, the eigenvalues of $-A$ are

$$\lambda_k(-A) = \frac{4}{h^2} \sin^2 \left(\frac{k\pi}{2m} \right), \quad k = 1, \dots, m-1.$$

In particular,

$$\lambda_{\min}(-A) = \frac{4}{h^2} \sin^2 \left(\frac{\pi}{2m} \right).$$

For large m , the quantity $\pi/(2m)$ is very small. Using the small-angle approximation

$$\sin \theta \approx \theta,$$

we obtain

$$\lambda_{\min}(-A) \approx \frac{4}{h^2} \left(\frac{\pi}{2m} \right)^2.$$

Since $h = L/m$,

$$(2.34) \quad \lambda_{\min}(-A) \approx \frac{\pi^2}{L^2}.$$

Recall that the spectral norm of a matrix is given by

$$\|A^{-1}\|_2 = \sqrt{\lambda_{\max}((A^{-1})^T A^{-1})}.$$

Since A is symmetric, A^{-1} is also symmetric. Therefore,

$$(A^{-1})^T A^{-1} = (A^{-1})^2.$$

Hence,

$$\|A^{-1}\|_2 = \sqrt{\lambda_{\max}((A^{-1})^2)}.$$

Now, if $\lambda_k(A)$ is an eigenvalue of A , then

$$\frac{1}{\lambda_k(A)}$$

is an eigenvalue of A^{-1} . Therefore,

$$\frac{1}{\lambda_k(A)^2}$$

is an eigenvalue of $(A^{-1})^2$. Thus,

$$\lambda_{\max}((A^{-1})^2) = \max_k \left| \frac{1}{\lambda_k(A)^2} \right|.$$

Since A is negative definite, the eigenvalues of $-A$ are positive. Therefore,

$$\|A^{-1}\|_2 = \frac{1}{\lambda_{\min}(-A)}.$$

From the previous subsection, the contraction condition is

$$\|A^{-1}\|_2 L_F < 1,$$

where

$$L_F = \frac{\lambda}{V_T} \cosh \left(\frac{M}{V_T} \right).$$

Substituting the spectral norm estimate gives

$$\frac{L_F}{\lambda_{\min}(-A)} < 1.$$

Equivalently,

$$L_F < \lambda_{\min}(-A).$$

Using (2.34), we obtain

$$L_F < \frac{\pi^2}{L^2}.$$

Therefore,

$$L^2 < \frac{\pi^2}{L_F},$$

and hence

$$L < \frac{\pi}{\sqrt{L_F}}.$$

Substituting the expression for L_F ,

$$L < \pi \sqrt{\frac{V_T}{\lambda \cosh(M/V_T)}}.$$

Recall from the intrinsic hyperbolic formulation that

$$\lambda = \frac{2qn_i}{\varepsilon}.$$

Substituting this into the threshold condition gives

$$L < \pi \sqrt{\frac{\varepsilon V_T}{2qn_i \cosh(M/V_T)}}.$$

The Debye screening length is defined by [VGK17, 06]

$$L_D = \sqrt{\frac{\varepsilon V_T}{2qn_i}}.$$

Therefore,

$$L < \frac{\pi L_D}{\sqrt{\cosh(M/V_T)}}.$$

Equivalently,

$$\frac{L}{L_D} < \frac{\pi}{\sqrt{\cosh(M/V_T)}}.$$

Since $\cosh(M/V_T) \geq 1$, we have

$$\frac{\pi}{\sqrt{\cosh(M/V_T)}} \leq \pi.$$

Also, since

$$3 < \pi < 3.2,$$

the best possible upper bound from this estimate is slightly larger than three Debye lengths. In the weakly nonlinear regime, where M/V_T is small and $\cosh(M/V_T) \approx 1$, the sufficient contraction condition becomes approximately

$$L < \pi L_D.$$

Thus, this analysis suggests that the fixed-point argument guarantees convergence when the device length is on the order of only a few Debye lengths.

As M/V_T increases, the quantity $\cosh(M/V_T)$ increases, so the upper bound on L/L_D decreases. Therefore, larger potential variations make the contraction condition more restrictive. If this sufficient bound is not satisfied, the analysis does not guarantee the convergence of the fixed-point iteration anymore. Nevertheless, this does not necessarily mean that the iteration diverges.

3. Numerical Analysis & Equilibrium Simulation

3.1. Numerical Solver Setup. In Chapter 2, the one-dimensional

Poisson–Boltzmann equation was discretized using finite differences and reformulated as a nonlinear algebraic system of the form

$$A\Phi = F[\Phi].$$

The associated fixed-point formulation, contraction properties, and spectral behavior of the discrete operator were also analyzed. We now turn to the computational implementation of this nonlinear system and construct iterative numerical schemes for obtaining numerical solutions. The computational domain is discretized using a uniform mesh

$$x_i = ih, \quad h = \frac{L}{m},$$

for $i = 0, \dots, m$. The discrete system used in the numerical implementation is

$$\begin{bmatrix} 1 & 0 & 0 & \cdots & 0 & 0 \\ 1 & -2 & 1 & \cdots & 0 & 0 \\ 0 & 1 & -2 & \ddots & 0 & 0 \\ \vdots & \vdots & \ddots & \ddots & 1 & 0 \\ 0 & 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & \cdots & 0 & 1 \end{bmatrix} \begin{bmatrix} \phi_0 \\ \phi_1 \\ \vdots \\ \phi_{m-1} \\ \phi_m \end{bmatrix} = h^2 \begin{bmatrix} -V_T \ln \left(\frac{|C(0)|}{n_i} \right) \\ \lambda \sinh \left(\frac{\phi_1}{V_T} \right) - \frac{q}{\varepsilon} C(x_1) \\ \vdots \\ \lambda \sinh \left(\frac{\phi_{m-1}}{V_T} \right) - \frac{q}{\varepsilon} C(x_{m-1}) \\ V_T \ln \left(\frac{C(L)}{n_i} \right) \end{bmatrix}.$$

This nonlinear system can be approached in several ways, which are analysed and simulated in the following subsections.

3.2. Standard Fixed–Point Iteration. The one-dimensional nonlinear

Poisson–Boltzmann equation may be solved using the fixed-point formulation derived in the previous subsection. Recall that the discrete nonlinear system was written in the form

$$\Phi = T[\Phi] = A^{-1}F[\Phi].$$

Starting from an initial guess

$$\Phi^{(0)},$$

the standard fixed-point iteration is defined by

$$\Phi^{(k+1)} = T[\Phi^{(k)}] = A^{-1}F[\Phi^{(k)}].$$

At each iteration, the nonlinear term is evaluated using the previous iterate, and the resulting linear system is solved to obtain an updated approximation of the electrostatic potential. The iteration is terminated when the absolute difference between successive iterates satisfies a prescribed tolerance. In the numerical experiments presented below, convergence is measured using the discrete L_∞ -norm:

$$\|\Phi^{(k+1)} - \Phi^{(k)}\|_\infty < \varepsilon,$$

where $\varepsilon > 0$ denotes a fixed numerical tolerance.

3.2.1. Limitation of Standard Fixed-Point Iteration. The standard fixed-point iteration provides a simple numerical approach for solving the nonlinear Poisson-Boltzmann equation. However, its convergence is highly sensitive to the strength of the nonlinear forcing.

To investigate this behavior, a sequence of numerical experiments was performed for a one-dimensional silicon p-n junction with symmetric doping profiles $N_A = N_D$. The convergence behavior for increasing doping magnitudes is summarized in Table 1. For each simulation, the iteration was initialized using the same linear initial guess and terminated when

$$\|\Phi^{(k+1)} - \Phi^{(k)}\|_\infty < 10^{-5}.$$

Doping	Converged	Iterations	Final Error	Status
1e+15	1	6	1.895e-06	Converged
2e+15	1	33	9.696e-06	Converged
3e+15	0	36	8.969e+43	Diverged/overflow
4e+15	0	4	7.617e+82	Diverged/overflow
5e+15	0	3	2.949e+66	Diverged/overflow
6e+15	0	3	NaN	Diverged/overflow
7e+15	0	3	NaN	Diverged/overflow
8e+15	0	3	NaN	Diverged/overflow
9e+15	0	3	NaN	Diverged/overflow
1e+16	0	3	NaN	Diverged/overflow

Table 1. Convergence behavior of the standard fixed-point iteration for increasing doping magnitudes in a symmetric silicon p-n junction.

Table 1 shows that the standard fixed-point iteration converges only for relatively mild doping regimes. For example, the method converges rapidly for doping magnitudes of 10^{15} cm^{-3} and $2 \times 10^{15} \text{ cm}^{-3}$, requiring only 6 and 33 iterations, respectively. However, beginning near $3 \times 10^{15} \text{ cm}^{-3}$, the iteration becomes unstable and produces extremely large numerical errors, eventually leading to floating-point overflow and undefined values. This behavior is consistent with the contraction mapping analysis developed in Chapter 2. As the doping magnitude increases, the nonlinear hyperbolic sine term grows rapidly, causing the fixed-point map to lose its contractive behavior. Consequently, the standard fixed-point iteration becomes unreliable for strongly nonlinear regimes.

These observations motivate the development of more robust iterative schemes. In the next subsection, we introduce a quasi-linearization approach for the nonlinear Poisson-Boltzmann equation, which significantly improves stability and convergence for higher doping regimes.

3.3. Quasi-linearization of Poisson-Boltzmann Equation. The numerical experiments in the previous subsection show that the standard fixed-point iteration

becomes unreliable as the doping magnitude increases. To improve stability, we now introduce a quasi-linearization method [VGK17]. Instead of evaluating the entire nonlinear term explicitly from the previous iterate, we linearize the nonlinear term around the current iterate and solve the resulting linear boundary-value problem at each step.

Starting from the hyperbolic form

$$\frac{d^2\phi}{dx^2} = \lambda \sinh\left(\frac{\phi}{V_T}\right) - \frac{q}{\varepsilon}C(x),$$

we linearize the nonlinear term around the current iterate $\phi^{(k)}$. Write

$$\phi^{(k+1)} = \phi^{(k)} + \delta.$$

Using a first-order Taylor expansion,

$$\sinh\left(\frac{\phi^{(k+1)}}{V_T}\right) \approx \sinh\left(\frac{\phi^{(k)}}{V_T}\right) + \frac{1}{V_T} \cosh\left(\frac{\phi^{(k)}}{V_T}\right) (\phi^{(k+1)} - \phi^{(k)}).$$

Substituting this into the Poisson-Boltzmann equation gives

$$\frac{d^2\phi^{(k+1)}}{dx^2} = \lambda \sinh\left(\frac{\phi^{(k)}}{V_T}\right) + \frac{\lambda}{V_T} \cosh\left(\frac{\phi^{(k)}}{V_T}\right) (\phi^{(k+1)} - \phi^{(k)}) - \frac{q}{\varepsilon}C(x).$$

Rearranging terms,

$$\frac{d^2\phi^{(k+1)}}{dx^2} - \frac{\lambda}{V_T} \cosh\left(\frac{\phi^{(k)}}{V_T}\right) \phi^{(k+1)} = \lambda \sinh\left(\frac{\phi^{(k)}}{V_T}\right) - \frac{q}{\varepsilon}C(x) - \frac{\lambda}{V_T} \cosh\left(\frac{\phi^{(k)}}{V_T}\right) \phi^{(k)}.$$

This equation is linear in the unknown $\phi^{(k+1)}$.

Using the central difference approximation at an interior node x_i , we obtain

$$\frac{\phi_{i-1}^{(k+1)} - 2\phi_i^{(k+1)} + \phi_{i+1}^{(k+1)}}{h^2} - \frac{\lambda}{V_T} \cosh\left(\frac{\phi_i^{(k)}}{V_T}\right) \phi_i^{(k+1)} = \lambda \sinh\left(\frac{\phi_i^{(k)}}{V_T}\right) - \frac{q}{\varepsilon}C_i - \frac{\lambda}{V_T} \cosh\left(\frac{\phi_i^{(k)}}{V_T}\right) \phi_i^{(k)}.$$

Multiplying by h^2 , the interior equation becomes

$$\phi_{i-1}^{(k+1)} - \left[2 + h^2 \frac{\lambda}{V_T} \cosh\left(\frac{\phi_i^{(k)}}{V_T}\right)\right] \phi_i^{(k+1)} + \phi_{i+1}^{(k+1)} = h^2 \lambda \sinh\left(\frac{\phi_i^{(k)}}{V_T}\right) - h^2 \frac{q}{\varepsilon}C_i - h^2 \frac{\lambda}{V_T} \cosh\left(\frac{\phi_i^{(k)}}{V_T}\right) \phi_i^{(k)}.$$

Thus, at each iteration, the quasi-linearization method solves a tridiagonal linear system

$$A^{(k)}\Phi^{(k+1)} = b^{(k)}.$$

The interior diagonal entries of $A^{(k)}$ are

$$-2 - h^2 \frac{\lambda}{V_T} \cosh \left(\frac{\phi_i^{(k)}}{V_T} \right),$$

while the off-diagonal entries remain 1.

Since $\cosh \left(\frac{\phi_i^{(k)}}{V_T} \right) > 0$, we have

$$\left| -2 - h^2 \frac{\lambda}{V_T} \cosh \left(\frac{\phi_i^{(k)}}{V_T} \right) \right| > |1| + |1| = 2.$$

Therefore, the matrix $A^{(k)}$ is strictly diagonally dominant. Compared to the standard fixed-point matrix, the additional hyperbolic cosine term strengthens the diagonal dominance of the linear system, which helps explain the improved numerical stability of the quasi-linearized iteration in strongly nonlinear regimes.

In the following numerical experiments, we apply this quasi-linearized scheme to simulate similar symmetric silicon p-n junction used in previous fixed-point experiments.

3.3.1. Convergence Behavior. To investigate the numerical behavior of the quasi-linearized method, the error between successive iterates was recorded throughout the nonlinear solve. Since the equilibrium Poisson-Boltzmann equation is strongly nonlinear, monitoring the error history provides insight into how efficiently the iteration approaches the numerical solution. Figure 1 shows the convergence history of the solver using a logarithmic scale on vertical axis.

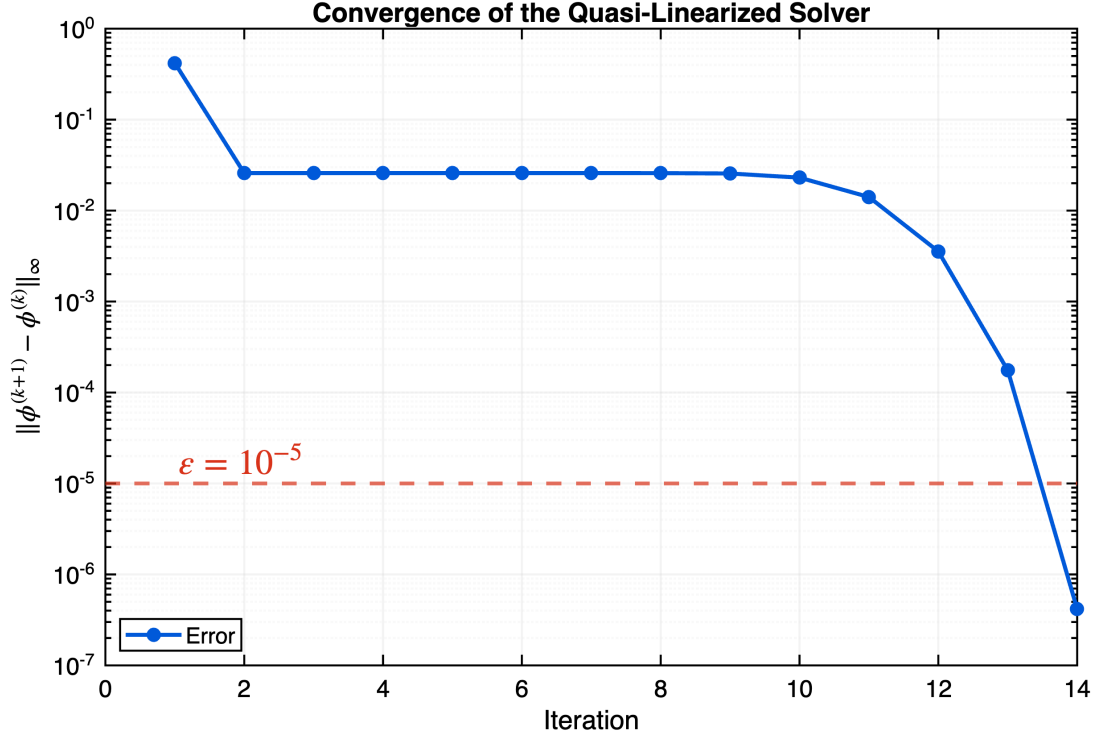


Figure 1. Convergence behavior of the quasi-linearized solver.

The convergence history exhibits two distinct phases. Following an initial reduction in the error from approximately 0.4×10^0 to 2.6×10^{-2} , the iteration enters a plateau region where the error remains nearly constant from approximately iteration 2 through iteration 10. Beyond iteration 10, the error begins to decrease rapidly and eventually crosses the prescribed tolerance of 10^{-5} at approximately the 14th iteration. The final error is approximately 4.16×10^{-6} , indicating successful convergence of the quasi-linearized solver.

Overall, the convergence history indicates that the quasi-linearized method successfully resolves the nonlinear Poisson-Boltzmann system while maintaining stable behavior throughout the iteration process.

3.3.2. Physics-Informed Initialization. In the previous simulation, the solver was initialized using a global linear interpolation between the boundary values. However,

the equilibrium p–n junction solution is nearly constant in the quasi–neutral regions and varies rapidly only near the depletion layer. Since

$$W_d \propto \frac{1}{\sqrt{N}},$$

the transition region becomes narrower as the doping magnitude increases. As a result, a global linear interpolation distributes variation across the entire device, whereas the true solution concentrates most of its variation near the junction. This mismatch can produce a relatively poor initial approximation and may require additional nonlinear iterations before the solver begins to accurately resolve the internal-layer structure.

Motivated by these observations, we initialize the solver using a piecewise profile consisting of a left plateau, a narrow transition region near the junction, and a right plateau. The junction is assumed to be located at the center of the device, and a small transition region is prescribed around this location. The left portion of the device is initialized using the left boundary potential ϕ_0 , while the right portion is initialized using the right boundary potential ϕ_m . Within the transition region, the two plateau values are connected using a simple linear interpolation scheme. Consequently, most of the variation in the initial guess is concentrated near the junction, while the outer regions remain nearly constant. Although this construction remains relatively simple, it captures the most important qualitative feature of the equilibrium solution, namely that the electrostatic potential is nearly constant away from the junction and changes rapidly only within a narrow transition layer. As a result, the initial iterate more closely resembles the expected solution profile than a global linear interpolation across the entire device.

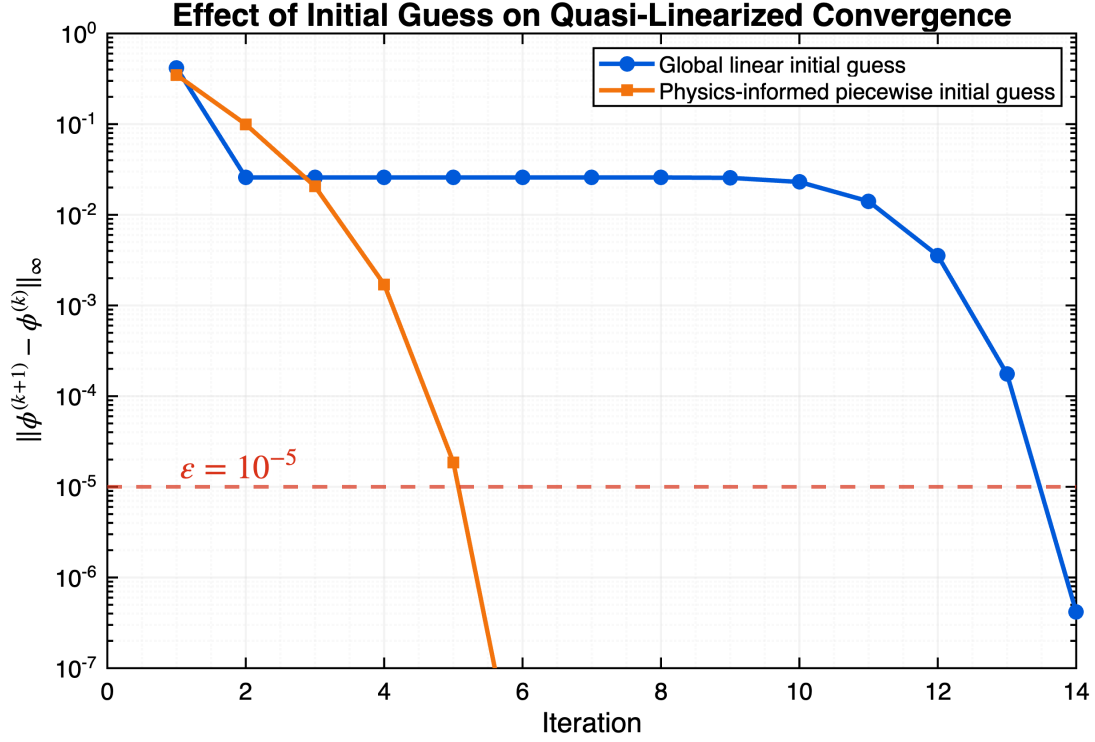


Figure 2. Initial Guess Comparison

Figure 2 compares the convergence histories obtained using a global linear initialization and the piecewise initialization described above. The piecewise profile lacks initial plateau phase and achieves the prescribed tolerance in significantly fewer iterations. In particular, the error level reached by the piecewise initialization after approximately (5) to (6) iterations is not attained by the global linear initialization until roughly iteration (13). This represents a noticeable reduction in the computational effort required to reach a comparable level of accuracy. The results demonstrate that incorporating qualitative features of the expected solution structure into the initial guess can significantly improve the efficiency of the nonlinear solve.

3.3.3. Electrostatic Potential & Electric Field. The electrostatic potential and electric field are shown below.

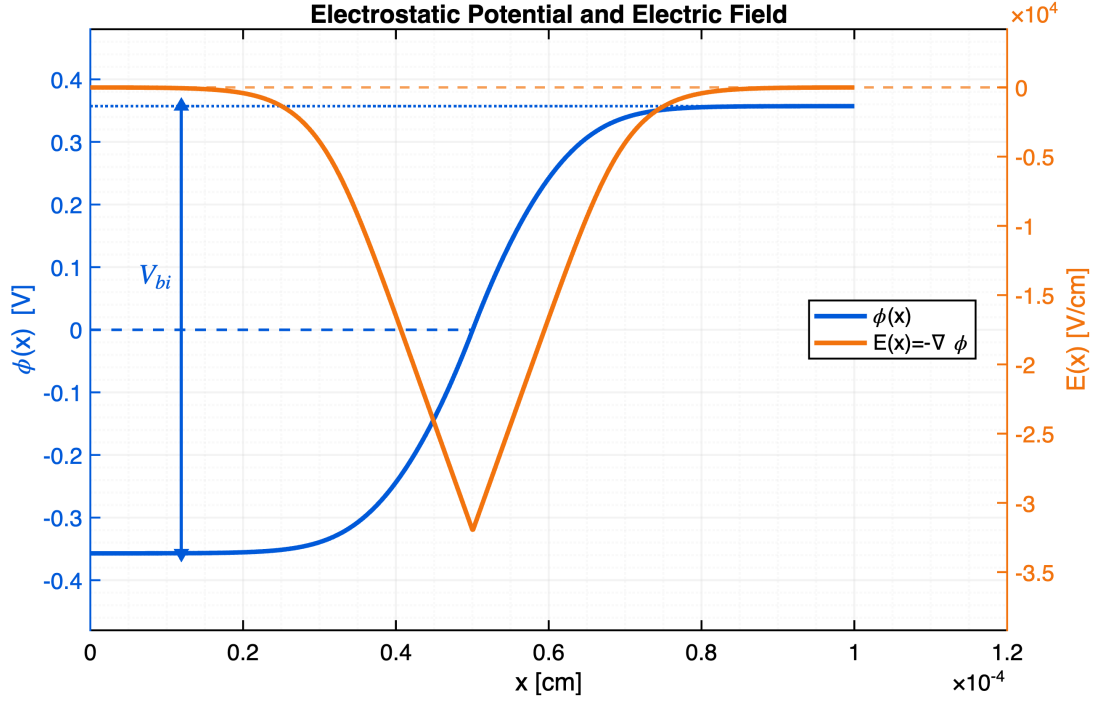


Figure 3. Electrostatic potential and electric field across the p–n junction.

The electric field is obtained from the electrostatic potential through

$$E = -\nabla\phi.$$

As shown in Figure 3, the electrostatic potential varies by approximately the built-in potential V_{bi} across the junction. Most of this variation occurs within a narrow region surrounding the junction, while the potential remains nearly constant throughout the quasi-neutral regions. Consequently, the electric field is concentrated primarily within the depletion region, where it attains its largest magnitude. Away from the junction, the field rapidly approaches zero, reflecting the nearly neutral behavior of the bulk semiconductor regions. These results are consistent with the classical equilibrium picture of a p–n junction, in which the built-in electric field is localized within the depletion layer and drives carrier separation across the junction.

3.3.4. Carrier Densities & Depletion Region. The carrier densities and depletion region are shown below.

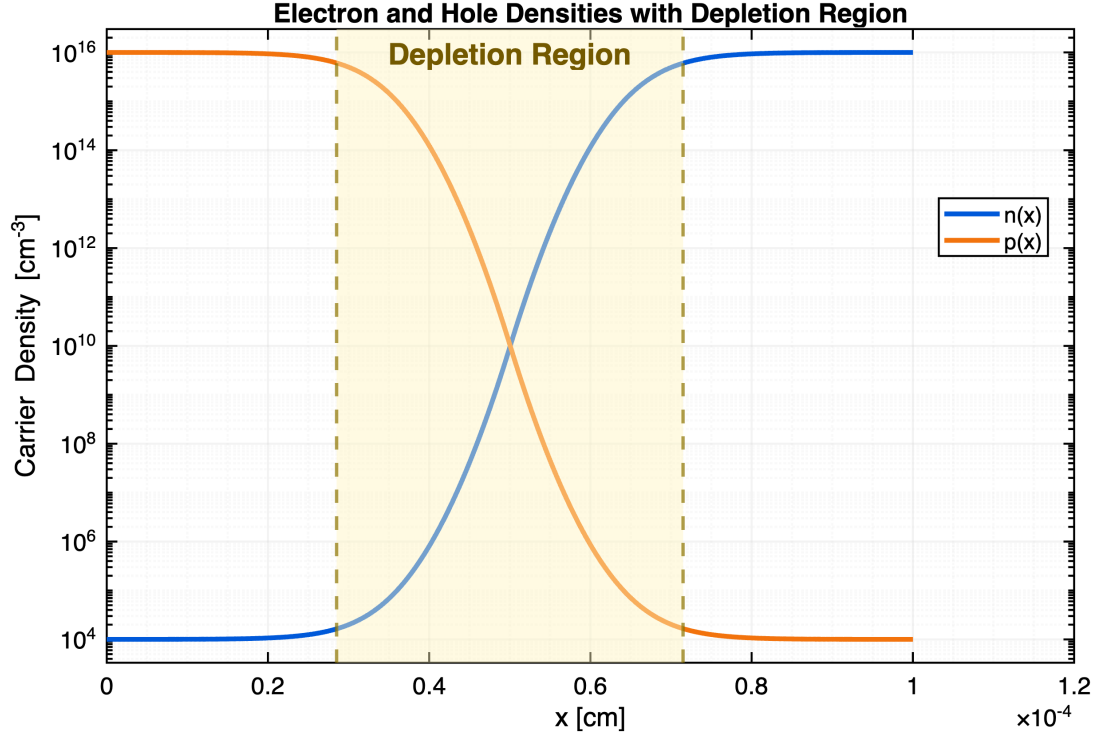


Figure 4. Electron and hole carrier densities with the depletion region highlighted.

The carrier density profiles shown in Figure 4 exhibit the expected equilibrium behavior of a p–n junction. On the p–side of the device, the hole concentration remains close to the acceptor doping level while the electron concentration is several orders of magnitude smaller. Similarly, on the n–side, the electron concentration dominates while the hole concentration becomes a minority carrier population. Near the junction, both carrier densities deviate significantly from their bulk values, giving rise to the depletion region highlighted in the figure.

The logarithmic scale reveals that the majority and minority carrier concentrations differ by several orders of magnitude throughout most of the device. The transition

between these regimes occurs primarily within the depletion region, where mobile charge carriers are depleted and the built-in electric field is strongest. These results are consistent with the classical equilibrium structure of a semiconductor p–n junction.

3.3.5. Doping Profile & Net Charge Density. The carrier densities and depletion region are shown below.

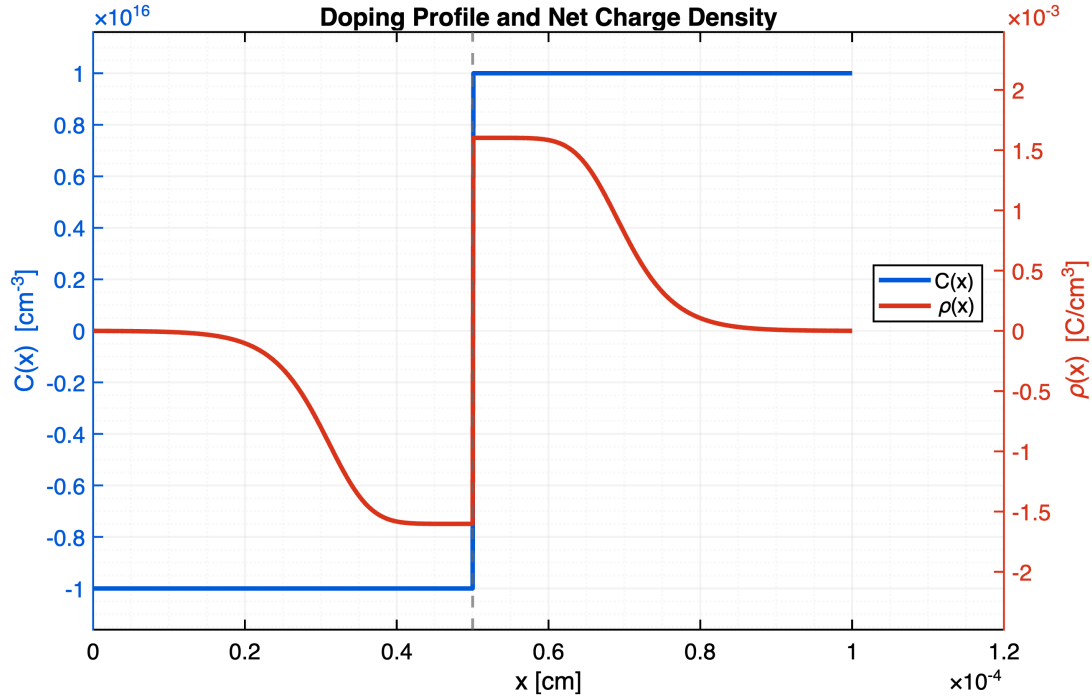


Figure 5. Doping profile and net charge density for the equilibrium p–n junction simulation.

The doping profile $C(x)$ specifies the location of p–n junction, with negative doping on the p–side and positive doping on the n–side of the device. As shown in Figure 5, the junction is located at the point where the doping profile changes sign.

Away from the junction, the mobile charge carriers nearly balance the doping profile, resulting in $\rho(x) \approx 0$. The net charge density changes sign across the junction and attains its largest magnitude within the depletion region. Outside this region, the charge density rapidly returns toward zero as charge neutrality is re-established.

These results are consistent with the classical equilibrium description of a p–n junction, where the depletion layer contains the uncompensated ionized dopants responsible for generating the built-in electric field.

3.4. Quasi-Linearization and Newton's Method. We now show that the quasi-linearized scheme derived previously is mathematically equivalent to Newton's method applied to the discrete nonlinear Poisson–Boltzmann system. To compare the quasi-linearized scheme with Newton's method, we return to the same discrete nonlinear system derived in the previous chapter. For each interior node $i = 1, \dots, m - 1$, the finite-difference equation can be written as

$$\phi_{i+1} - 2\phi_i + \phi_{i-1} - h^2 \lambda \sinh\left(\frac{\phi_i}{V_T}\right) + h^2 \frac{q}{\varepsilon} C_i = 0$$

we define,

$$F_i(\Phi) = \phi_{i+1} - 2\phi_i + \phi_{i-1} - h^2 \lambda \sinh\left(\frac{\phi_i}{V_T}\right) + h^2 \frac{q}{\varepsilon} C_i = 0$$

The boundary rows are enforced as:

$$F_0(\Phi) = \phi_0 - V_T \ln\left(\frac{n_i}{|C_1|}\right),$$

and

$$F_m(\Phi) = \phi_m - V_T \ln\left(\frac{C_m}{n_i}\right).$$

Newton's method solves the nonlinear system using the iteration

$$J(\Phi^{(k)}) \Delta \Phi^{(k)} = -F(\Phi^{(k)}),$$

with update

$$\Phi^{(k+1)} = \Phi^{(k)} + \Delta \Phi^{(k)}.$$

Substituting

$$\Delta \Phi^{(k)} = \Phi^{(k+1)} - \Phi^{(k)},$$

gives

$$J(\Phi^{(k)}) (\Phi^{(k+1)} - \Phi^{(k)}) = -F(\Phi^{(k)}),$$

which can be rearranged as

$$J(\Phi^{(k)})\Phi^{(k+1)} = J(\Phi^{(k)})\Phi^{(k)} - F(\Phi^{(k)}).$$

Since the nonlinear residual depends only on

$$\phi_{i-1}, \quad \phi_i, \quad \phi_{i+1},$$

the Jacobian matrix is tridiagonal. The Jacobian entries are

$$\frac{\partial F_i}{\partial \phi_i} = -2 - h^2 \frac{\lambda}{V_T} \cosh \left(\frac{\phi_i}{V_T} \right),$$

and

$$\frac{\partial F_i}{\partial \phi_{i-1}} = 1, \quad \frac{\partial F_i}{\partial \phi_{i+1}} = 1.$$

Thus, the Newton Jacobian takes the form

$$J(\Phi^{(k)}) = \begin{bmatrix} 1 & 0 & & & 0 \\ 1 & d_1^{(k)} & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & 1 & d_{m-1}^{(k)} & 1 \\ 0 & & & 0 & 1 \end{bmatrix},$$

where

$$d_i^{(k)} = -2 - h^2 \frac{\lambda}{V_T} \cosh \left(\frac{\phi_i^{(k)}}{V_T} \right).$$

Comparing the Newton Jacobian matrix with the matrix obtained in the quasi-linearized formulation shows that the two matrices are identical.

For the interior rows,

$$\left(J(\Phi^{(k)})\Phi^{(k)} - F(\Phi^{(k)}) \right)_i$$

becomes

$$\phi_{i-1}^{(k)} + d_i^{(k)} \phi_i^{(k)} + \phi_{i+1}^{(k)} - \left[\phi_{i+1}^{(k)} - 2\phi_i^{(k)} + \phi_{i-1}^{(k)} - h^2 \lambda \sinh \left(\frac{\phi_i^{(k)}}{V_T} \right) + h^2 \frac{q}{\varepsilon} C_i \right].$$

The discrete Laplacian terms cancel exactly, leaving

$$h^2 \lambda \sinh \left(\frac{\phi_i^{(k)}}{V_T} \right) - h^2 \frac{q}{\varepsilon} C_i - h^2 \frac{\lambda}{V_T} \cosh \left(\frac{\phi_i^{(k)}}{V_T} \right) \phi_i^{(k)},$$

which is precisely the right-hand side obtained in the quasi-linearized formulation.

Therefore, the quasi-linearized scheme is not merely Newton-type, but exactly Newton's method applied to the discrete nonlinear Poisson-Boltzmann system.

3.5. Quasi-Fermi Levels and Slotboom Variables.

3.5.1. Drift–Diffusion Current Formulation. We begin with the standard drift–diffusion current densities for electrons and holes:

$$J_n = q\mu_n nE + qD_n \nabla n,$$

$$J_p = q\mu_p pE - qD_p \nabla p,$$

where q is the elementary charge, μ_n and μ_p are the carrier mobilities, D_n and D_p are the diffusion coefficients, and E denotes the electric field.

Using the electrostatic relation

$$E = -\nabla\phi,$$

the current densities become

$$J_n = -q\mu_n n \nabla\phi + qD_n \nabla n,$$

$$J_p = -q\mu_p p \nabla\phi - qD_p \nabla p.$$

Applying the Einstein relations

$$D_n = \mu_n V_T, \quad D_p = \mu_p V_T,$$

where

$$V_T = \frac{k_B T}{q}$$

is the thermal voltage, we obtain

$$J_n = -q\mu_n n \nabla\phi + q\mu_n V_T \nabla n,$$

$$J_p = -q\mu_p p \nabla\phi - q\mu_p V_T \nabla p.$$

Factoring out the mobility terms gives

$$J_n = q\mu_n (-n \nabla\phi + V_T \nabla n),$$

$$J_p = -q\mu_p (p \nabla\phi + V_T \nabla p).$$

Therefore, the drift–diffusion currents may be written in the form

$$J_n = q\mu_n (-n \nabla\phi + V_T \nabla n),$$

$$J_p = -q\mu_p (p \nabla\phi + V_T \nabla p).$$

At this stage, both current densities contain a mixture of drift and diffusion contributions. For electrons, the transport mechanism appears through the combination

$$-n\nabla\phi + V_T\nabla n,$$

while for holes it appears as

$$p\nabla\phi + V_T\nabla p.$$

Although these formulations are physically correct, the drift and diffusion terms remain coupled together. This coupling is often inconvenient both analytically and numerically. To obtain a cleaner structure, it is useful to introduce alternative variables that rewrite the currents in a more compact form.

Two standard reformulations are commonly used:

- Quasi-Fermi potentials,
- Slotboom variables.

Both reformulations will be derived in the following sub-sections.

3.5.2. Quasi-Fermi Potentials. We begin with the electron current density

$$J_n = q\mu_n (-n\nabla\phi + V_T\nabla n).$$

Observe that

$$\nabla(\ln n) = \frac{1}{n}\nabla n,$$

so that

$$\nabla n = n\nabla(\ln n).$$

Substituting this identity into the electron current gives

$$J_n = q\mu_n (-n\nabla\phi + V_T n\nabla(\ln n)).$$

Factoring out n yields

$$J_n = q\mu_n n (-\nabla\phi + V_T\nabla(\ln n)).$$

We now define the electron quasi-Fermi potential by

$$\phi_n := \phi - V_T \ln\left(\frac{n}{n_i}\right),$$

where n_i denotes the intrinsic carrier concentration. The factor n_i is introduced so that the logarithm remains dimensionless.

Taking the gradient gives

$$\nabla \phi_n = \nabla \phi - V_T \nabla \left(\ln \left(\frac{n}{n_i} \right) \right).$$

Since n_i is constant,

$$\nabla \left(\ln \left(\frac{n}{n_i} \right) \right) = \nabla (\ln n),$$

and therefore

$$\nabla \phi_n = \nabla \phi - V_T \nabla (\ln n).$$

Multiplying by -1 gives

$$-\nabla \phi_n = -\nabla \phi + V_T \nabla (\ln n),$$

which is exactly the quantity appearing in the electron current density. Hence

$$J_n = -q\mu_n n \nabla \phi_n.$$

We now repeat the same procedure for holes. Starting from

$$J_p = -q\mu_p (p \nabla \phi + V_T \nabla p),$$

observe that

$$\nabla p = p \nabla (\ln p).$$

Substituting into the hole current gives

$$J_p = -q\mu_p (p \nabla \phi + V_T p \nabla (\ln p)).$$

Factoring out p yields

$$J_p = -q\mu_p p (\nabla \phi + V_T \nabla (\ln p)).$$

Define the hole quasi-Fermi potential by

$$\phi_p := \phi + V_T \ln \left(\frac{p}{n_i} \right).$$

Taking the gradient gives

$$\nabla \phi_p = \nabla \phi + V_T \nabla (\ln p),$$

and therefore

$$J_p = -q\mu_p p \nabla \phi_p.$$

Thus the drift–diffusion currents may be rewritten in the compact form

$$J_n = -q\mu_n n \nabla \phi_n,$$

$$J_p = -q\mu_p p \nabla \phi_p.$$

These expressions show that electron and hole transport are driven by gradients of the corresponding quasi-Fermi potentials [VGK17]. Instead of viewing carrier transport as a mixture of drift and diffusion, one may interpret it as transport induced by non-flat quasi-Fermi levels.

At thermodynamic equilibrium,

$$\nabla \phi_n = 0, \quad \nabla \phi_p = 0,$$

so that both currents vanish:

$$J_n = 0, \quad J_p = 0.$$

In one space dimension, the quasi-Fermi current densities reduce to

$$J_n = -q\mu_n n \frac{d\phi_n}{dx},$$

$$J_p = -q\mu_p p \frac{d\phi_p}{dx}.$$

The quasi-Fermi potentials are given by

$$\phi_n = \phi - V_T \ln \left(\frac{n}{n_i} \right),$$

$$\phi_p = \phi + V_T \ln \left(\frac{p}{n_i} \right).$$

Rearranging these expressions gives

$$\ln \left(\frac{n}{n_i} \right) = \frac{\phi - \phi_n}{V_T},$$

$$\ln \left(\frac{p}{n_i} \right) = \frac{\phi_p - \phi}{V_T}.$$

Exponentiating yields

$$n = n_i \exp \left(\frac{\phi - \phi_n}{V_T} \right),$$

$$p = n_i \exp \left(\frac{\phi_p - \phi}{V_T} \right).$$

Therefore, the carrier densities are determined by the difference between the electrostatic potential and the corresponding quasi-Fermi potentials.

3.5.3. Slotboom Variables. While the quasi-Fermi formulation rewrites the current densities in terms of gradients of quasi-Fermi potentials, another important reformulation is obtained through the introduction of the Slotboom variables. These variables transform the drift–diffusion currents into a divergence form that is particularly convenient for numerical discretization [VGK17].

Starting from the electron drift–diffusion current,

$$J_n = q\mu_n (-n\nabla\phi + V_T\nabla n),$$

we seek a transformation that rewrites the current as a coefficient multiplied by the gradient of a new variable.

Define the electron Slotboom variable by

$$u := ne^{-\phi/V_T}.$$

Applying the product rule,

$$\nabla u = \nabla (ne^{-\phi/V_T}),$$

gives

$$\nabla u = e^{-\phi/V_T}\nabla n + n\nabla (e^{-\phi/V_T}).$$

Since

$$\nabla (e^{-\phi/V_T}) = -\frac{1}{V_T}e^{-\phi/V_T}\nabla\phi,$$

we obtain

$$\nabla u = e^{-\phi/V_T} \left(\nabla n - \frac{n}{V_T} \nabla\phi \right).$$

Multiplying both sides by $q\mu_n V_T e^{\phi/V_T}$ yields

$$q\mu_n V_T e^{\phi/V_T} \nabla u = q\mu_n (V_T \nabla n - n \nabla\phi).$$

Therefore,

$$J_n = q\mu_n V_T e^{\phi/V_T} \nabla u.$$

Using the Einstein relation

$$D_n = \mu_n V_T,$$

the electron current becomes

$$J_n = qD_n e^{\phi/V_T} \nabla u.$$

A completely analogous derivation can be carried out for holes. Starting from

$$J_p = -q\mu_p (p\nabla\phi + V_T\nabla p),$$

define the hole Slotboom variable by

$$v := pe^{\phi/V_T}.$$

Differentiating,

$$\nabla v = \nabla (pe^{\phi/V_T}),$$

gives

$$\nabla v = e^{\phi/V_T} \nabla p + p \nabla (e^{\phi/V_T}).$$

Since

$$\nabla (e^{\phi/V_T}) = \frac{1}{V_T} e^{\phi/V_T} \nabla \phi,$$

we obtain

$$\nabla v = e^{\phi/V_T} \left(\nabla p + \frac{p}{V_T} \nabla \phi \right).$$

Multiplying by $-q\mu_p V_T e^{-\phi/V_T}$,

$$-q\mu_p V_T e^{-\phi/V_T} \nabla v = -q\mu_p (V_T \nabla p + p \nabla \phi).$$

Hence,

$$J_p = -q\mu_p V_T e^{-\phi/V_T} \nabla v.$$

Using

$$D_p = \mu_p V_T,$$

the hole current becomes

$$J_p = -qD_p e^{-\phi/V_T} \nabla v.$$

Therefore, the Slotboom formulation of the drift-diffusion currents is

$$J_n = qD_n e^{\phi/V_T} \nabla u, \quad J_p = -qD_p e^{-\phi/V_T} \nabla v.$$

The main advantage of this formulation is that the drift and diffusion terms are no longer separated. Instead, they are absorbed into weighted gradient operators involving

the Slotboom variables u and v . This structure plays a central role in the derivation of the Scharfetter–Gummel discretization developed in the next section.

3.6. Scharfetter–Gummel Discretization. The Slotboom formulation provides a convenient starting point for constructing exponentially fitted numerical fluxes. In semiconductor transport problems, standard finite difference discretizations often perform poorly in regions with strong electric fields or sharp boundary layers. The Scharfetter–Gummel discretization overcomes this difficulty by analytically integrating the drift–diffusion equation over each mesh cell [SG69].

We begin with the electron current written in Slotboom form:

$$J_n = qD_n e^{\phi/V_T} \nabla u,$$

where

$$u = n e^{-\phi/V_T}.$$

In one spatial dimension,

$$J_n = qD_n e^{\phi/V_T} \frac{du}{dx}.$$

Consider a mesh cell

$$[x_i, x_{i+1}],$$

with mesh spacing

$$h_i = x_{i+1} - x_i.$$

The Scharfetter–Gummel derivation assumes the following:

- (1) the current J_n is constant on the cell,
- (2) the electrostatic potential $\phi(x)$ varies linearly on the cell.

Under the linear potential assumption,

$$\phi(x) = \phi_i + \frac{\phi_{i+1} - \phi_i}{h_i} (x - x_i).$$

Therefore the electric field is constant on the cell:

$$E(x) = -\frac{d\phi}{dx} = -\frac{\phi_{i+1} - \phi_i}{h_i}.$$

Since J_n is assumed constant,

$$\frac{du}{dx} = \frac{J_n}{qD_n} e^{-\phi/V_T}.$$

Integrating over the cell gives

$$u_{i+1} - u_i = \frac{J_n}{qD_n} \int_{x_i}^{x_{i+1}} e^{-\phi(x)/V_T} dx.$$

Using the linear form of $\phi(x)$,

$$\phi(x) = \phi_i + \frac{\phi_{i+1} - \phi_i}{h_i} (x - x_i),$$

the exponential becomes

$$e^{-\phi(x)/V_T} = e^{-\phi_i/V_T} \exp\left(-\frac{\phi_{i+1} - \phi_i}{V_T h_i} (x - x_i)\right).$$

Introduce the dimensionless cell potential jump

$$\psi_i := \frac{\phi_{i+1} - \phi_i}{V_T}.$$

Then

$$e^{-\phi(x)/V_T} = e^{-\phi_i/V_T} e^{-\psi_i(x-x_i)/h_i}.$$

Hence,

$$\int_{x_i}^{x_{i+1}} e^{-\phi(x)/V_T} dx = h_i e^{-\phi_i/V_T} \frac{1 - e^{-\psi_i}}{\psi_i}.$$

Substituting into the integrated current relation yields

$$u_{i+1} - u_i = \frac{J_n h_i}{qD_n} e^{-\phi_i/V_T} \frac{1 - e^{-\psi_i}}{\psi_i}.$$

Recall that

$$u_i = n_i e^{-\phi_i/V_T}, \quad u_{i+1} = n_{i+1} e^{-\phi_{i+1}/V_T}.$$

After substitution and simplification,

$$J_{n,i+\frac{1}{2}} = \frac{qD_n}{h_i} [n_{i+1} B(\psi_i) - n_i B(-\psi_i)],$$

where the Bernoulli function is defined by

$$B(\psi) = \frac{\psi}{e^\psi - 1}.$$

The same derivation for holes gives

$$J_{p,i+\frac{1}{2}} = \frac{qD_p}{h_i} [p_i B(\psi_i) - p_{i+1} B(-\psi_i)].$$

The index

$$i + \frac{1}{2}$$

denotes the interface between adjacent mesh nodes x_i and x_{i+1} . The numerical flux is therefore associated with the cell edge rather than the node itself. One of the most important properties of the Scharfetter–Gummel discretization is that it exactly preserves local equilibrium.

The Scharfetter–Gummel discretization can be understood by looking at the drift–diffusion equation locally on a single mesh interval $[x_i, x_{i+1}]$. For the electron current, we write

$$J_n = q\mu_n n E + qD_n \frac{dn}{dx},$$

where $E = -d\phi/dx$. Using the Einstein relation $D_n = \mu_n V_T$, this becomes

$$J_n = q\mu_n \left(-n \frac{d\phi}{dx} + V_T \frac{dn}{dx} \right).$$

Let

$$\psi = \frac{\phi}{V_T}.$$

Then the current equation can be written as

$$\frac{dn}{dx} - n \frac{d\psi}{dx} = \frac{J_n}{qD_n}.$$

On each mesh interval, the Scharfetter–Gummel derivation assumes that the current J_n is constant and that the electrostatic potential varies linearly. Therefore $d\psi/dx$ is constant on $[x_i, x_{i+1}]$, and the carrier density satisfies a first-order linear differential equation with constant coefficients, which can be solved by finding Integrating Factor. This implies that the local carrier density is not linear. Instead, it has the form

$$n(x) = C_1 e^{\psi(x)} + C_2,$$

where C_1 and C_2 are constants determined by the endpoint values n_i and n_{i+1} .

Thus, the Scharfetter–Gummel method is not based on a linear approximation of the carrier density. Instead, it exactly integrates the local drift–diffusion equation under the assumptions of constant current and linear electrostatic potential. This hidden

exponential structure is one of the main reasons why the method remains stable and accurate in drift-dominated semiconductor transport regimes. Unlike standard central difference discretizations, the Scharfetter–Gummel flux remains stable and accurate even in strongly drift-dominated regimes.

3.7. Comparison of Quasi-Fermi and Scharfetter–Gummel Formulations.

The quasi-Fermi, Slotboom, and Scharfetter–Gummel formulations are closely related descriptions of the same transport structure. At the continuous level, these formulations are analytically equivalent representations of the drift–diffusion equations. The differences arise only after discretization.

The Scharfetter–Gummel electron flux on a mesh cell $[x_i, x_{i+1}]$ is

$$J_{n,i+\frac{1}{2}}^{SG} = \frac{qD_n}{h} [n_{i+1}B(\psi) - n_iB(-\psi)],$$

where

$$\psi = \frac{\phi_{i+1} - \phi_i}{V_T}, \quad B(\psi) = \frac{\psi}{e^\psi - 1}.$$

The Slotboom formulation rewrites the current as

$$J_n = qD_n e^{\phi/V_T} \frac{du}{dx}, \quad u = ne^{-\phi/V_T},$$

while the quasi-Fermi formulation becomes

$$J_n = -q\mu_n n \frac{d\phi_n}{dx}.$$

At the continuous level these formulations describe the same transport law written in different variables. The nontrivial comparison begins only after introducing a discrete approximation.

3.7.1. Discrete Quasi-Fermi Flux.

Starting from the quasi-Fermi flux

$$J_n = -q\mu_n n \frac{d\phi_n}{dx},$$

we discretize the quasi-Fermi gradient on the mesh cell using a finite difference approximation.

Define the logarithmic mean density

$$L(n_i, n_{i+1}) = \frac{n_{i+1} - n_i}{\ln n_{i+1} - \ln n_i}.$$

Replacing the cell density by its logarithmic mean gives

$$J_{n,i+\frac{1}{2}}^{QF} = -q\mu_n L(n_i, n_{i+1}) \frac{\phi_{n,i+1} - \phi_{n,i}}{h}.$$

Introduce the notation

$$r := \frac{n_{i+1}}{n_i},$$

together with the nonequilibrium parameter

$$\eta := \ln r - \psi.$$

Using the quasi-Fermi relation

$$\phi_{n,i+1} - \phi_{n,i} = -V_T \eta,$$

and the Einstein relation

$$D_n = \mu_n V_T,$$

we obtain

$$J_{n,i+\frac{1}{2}}^{QF} = \frac{qD_n}{h} L(n_i, n_{i+1}) \eta.$$

Since

$$L(n_i, n_{i+1}) = n_i \frac{r - 1}{\ln r},$$

the flux becomes

$$J_{n,i+\frac{1}{2}}^{QF} = \frac{qD_n}{h} n_i \frac{r - 1}{\ln r} \eta.$$

Using

$$r = e^{\psi+\eta}, \quad \ln r = \psi + \eta,$$

gives

$$J_{n,i+\frac{1}{2}}^{QF} = \frac{qD_n}{h} n_i \frac{e^{\psi+\eta} - 1}{\psi + \eta} \eta.$$

3.7.2. Exact Equilibrium Condition. Both the Scharfetter–Gummel and quasi-Fermi fluxes vanish under the same local equilibrium condition

$$\eta = 0.$$

Since

$$\eta = \ln r - \psi,$$

this condition is equivalent to

$$r = e^\psi.$$

Therefore,

$$J_{n,i+\frac{1}{2}}^{SG} = 0 \quad \Longleftrightarrow \quad J_{n,i+\frac{1}{2}}^{QF} = 0.$$

Thus both discretizations preserve the same exact local equilibrium relation.

3.7.3. Same Sign Structure. For all real ψ ,

$$B(\psi) > 0, \quad B(-\psi) > 0.$$

Also,

$$e^\eta - 1$$

has the same sign as η . Consequently, the Scharfetter–Gummel flux has the same sign as η .

The logarithmic-mean quasi-Fermi flux is also proportional to η with a positive prefactor. Hence

$$\text{sign} \left(J_{n,i+\frac{1}{2}}^{SG} \right) = \text{sign} \left(J_{n,i+\frac{1}{2}}^{QF} \right) = \text{sign}(\eta).$$

Therefore, although the two fluxes may differ in magnitude, they drive transport in the same direction relative to equilibrium.

3.7.4. Asymptotic Equivalence Regime. The natural comparison regime is

$$|\eta| \ll 1, \quad |\psi| \ll 1.$$

The condition

$$|\eta| \ll 1$$

corresponds to near local equilibrium, while

$$|\psi| \ll 1$$

corresponds to weak electric field variation across the mesh cell.

We now compare the logarithmic-mean quasi-Fermi flux with the Scharfetter–Gummel flux in this regime.

From the Scharfetter–Gummel formula,

$$J_{n,i+\frac{1}{2}}^{SG} = \frac{qD_n}{h} n_i B(-\psi) (e^\eta - 1).$$

Therefore,

$$\frac{J_{n,i+\frac{1}{2}}^{QF}}{J_{n,i+\frac{1}{2}}^{SG}} = \frac{e^{\psi+\eta} - 1}{(\psi + \eta) B(-\psi) (e^\eta - 1)} \eta.$$

Now consider the limit

$$\eta \rightarrow 0.$$

Using

$$e^\eta = 1 + \eta + O(\eta^2),$$

we obtain

$$e^\eta - 1 = \eta + O(\eta^2).$$

Similarly,

$$e^{\psi+\eta} = e^\psi e^\eta = e^\psi (1 + \eta + O(\eta^2)).$$

Thus,

$$e^{\psi+\eta} - 1 = (e^\psi - 1) + e^\psi \eta + O(\eta^2).$$

Substituting into the ratio gives

$$\frac{J_{n,i+\frac{1}{2}}^{QF}}{J_{n,i+\frac{1}{2}}^{SG}} \sim \frac{e^\psi - 1}{\psi B(-\psi)} \quad (\eta \rightarrow 0).$$

Using the Bernoulli definition,

$$B(-\psi) = \frac{-\psi}{e^{-\psi} - 1} = \frac{\psi e^\psi}{e^\psi - 1},$$

we obtain

$$\frac{J_{n,i+\frac{1}{2}}^{QF}}{J_{n,i+\frac{1}{2}}^{SG}} \sim \frac{(e^\psi - 1)^2}{\psi^2 e^\psi}.$$

Now rewrite the numerator as

$$(e^\psi - 1)^2 = e^\psi (e^{\psi/2} - e^{-\psi/2})^2.$$

Therefore,

$$\frac{J_{n,i+\frac{1}{2}}^{QF}}{J_{n,i+\frac{1}{2}}^{SG}} \sim \frac{(e^{\psi/2} - e^{-\psi/2})^2}{\psi^2}.$$

Since

$$e^{\psi/2} - e^{-\psi/2} = 2 \sinh(\psi/2),$$

we arrive at

$$\frac{J_{n,i+\frac{1}{2}}^{QF}}{J_{n,i+\frac{1}{2}}^{SG}} \sim \left(\frac{\sinh(\psi/2)}{\psi/2} \right)^2.$$

We now expand the hyperbolic sine term. Using the Taylor expansion

$$\sinh(x) = x + \frac{x^3}{6} + O(x^5),$$

gives

$$\frac{\sinh(\psi/2)}{\psi/2} = 1 + \frac{\psi^2}{24} + O(\psi^4).$$

Squaring,

$$\left(\frac{\sinh(\psi/2)}{\psi/2} \right)^2 = 1 + \frac{\psi^2}{12} + O(\psi^4).$$

Therefore,

$$J_{n,i+\frac{1}{2}}^{QF} = J_{n,i+\frac{1}{2}}^{SG} \left(1 + \frac{\psi^2}{12} + O(\psi^4) \right)$$

in the near-equilibrium and weak-field regime.

Thus the logarithmic-mean quasi-Fermi discretization is asymptotically equivalent to the Scharfetter–Gummel flux.

3.7.5. Error Statement. The difference between the logarithmic-mean quasi-Fermi flux and the Scharfetter–Gummel flux can now be expressed asymptotically in terms of the parameters ψ and η .

Near local equilibrium and under weak electric field variation,

$$J_{n,i+\frac{1}{2}}^{QF} - J_{n,i+\frac{1}{2}}^{SG} = \frac{qD_n}{h} n_i O(\psi^2 \eta + \eta^2).$$

Hence the quasi-Fermi discretization agrees with the Scharfetter–Gummel flux to leading order, while the first mismatch appears only through higher-order nonequilibrium corrections.

3.7.6. Conclusion of the Comparison. There are two distinct levels in this comparison.

At the continuous level, the quasi-Fermi, Slotboom, and Scharfetter–Gummel formulations describe the same transport equation written in different mathematical languages.

At the discrete level, however, equality is no longer automatic. Once the quasi-Fermi gradient and carrier density are discretized, the resulting numerical flux depends on the particular approximation introduced at the cell level.

The Scharfetter–Gummel formulation implicitly preserves the exponential intra-cell structure generated by the drift–diffusion equation under constant-current and linear-potential assumptions. In contrast, quasi-Fermi discretizations depend on the chosen approximation of the carrier density within the cell.

Nevertheless, the logarithmic-mean quasi-Fermi discretization preserves several important structural properties of the Scharfetter–Gummel scheme:

- the same exact equilibrium condition,
- the same sign structure,
- asymptotic equivalence near equilibrium,
- agreement to leading order under weak electric fields.

Therefore, the Scharfetter–Gummel and quasi-Fermi formulations should not be viewed as fundamentally unrelated competing approaches. Rather, they are closely connected discretizations whose differences emerge primarily from the discrete approximation introduced at each cell level.

4. Non-Equilibrium Drift–Diffusion Analysis and Simulation

4.1. Non-Equilibrium Drift–Diffusion System. In Section 2.2, the full drift–diffusion system was introduced and subsequently reduced to an equilibrium model by assuming thermodynamic equilibrium together with vanishing recombination-generation effects. Under these assumptions,

$$\frac{\partial n}{\partial t} = \frac{\partial p}{\partial t} = 0, \quad R = 0,$$

which led to the equilibrium Poisson–Boltzmann formulation studied in the previous chapters.

We now return to the full non-equilibrium drift–diffusion system in order to model carrier transport under applied bias. In the non-equilibrium regime, electron and hole currents are no longer balanced by equilibrium drift-diffusion cancellation, and recombination effects must be included explicitly [VGK17, 06].

The steady-state non-equilibrium drift–diffusion–Poisson system is given by

$$-\nabla \cdot (\epsilon \nabla \phi) = q(p - n + C),$$

together with the carrier continuity equations

$$\nabla \cdot J_n = qR,$$

$$\nabla \cdot J_p = -qR.$$

The electron and hole current densities are

$$J_n = q\mu_n n E + qD_n \nabla n,$$

$$J_p = q\mu_p p E - qD_p \nabla p,$$

where

$$E = -\nabla \phi.$$

In contrast to the equilibrium case, the carrier currents are now nonzero and are coupled through both the electrostatic potential and the recombination term.

To model recombination effects, the Shockley–Read–Hall (SRH) recombination model [SR52, Hal52] is used:

$$R = \frac{np - n_i^2}{\tau_p(n + n_1) + \tau_n(p + p_1)}.$$

Here, τ_n and τ_p denote the electron and hole carrier lifetimes, while n_1 and p_1 are the SRH reference densities associated with the trap level.

Under applied bias, the electron and hole quasi-Fermi levels separate, producing nonzero carrier transport across the device. As a result, the coupled drift–diffusion system becomes significantly more nonlinear and numerically stiff than the equilibrium Poisson–Boltzmann model studied previously.

4.2. Boundary Conditions Under Applied Bias. The equilibrium built-in potential is given by

$$V_{bi} = V_T \ln \left(\frac{N_A N_D}{n_i^2} \right).$$

Under an applied voltage V_{app} , the effective barrier becomes

$$V_{eff} = V_{bi} - V_{app}.$$

The equilibrium contact potentials are

$$\phi_L^{eq} = V_T \ln \left(\frac{n_i}{N_A} \right), \quad \phi_R^{eq} = V_T \ln \left(\frac{N_D}{n_i} \right).$$

The applied voltage is incorporated through the potential boundary conditions

$$\phi(0) = \phi_L^{eq} + \frac{V_{app}}{2}, \quad \phi(L) = \phi_R^{eq} - \frac{V_{app}}{2}.$$

The carrier densities satisfy ohmic contact conditions:

$$\begin{aligned} n(0) &= \frac{n_i^2}{N_A}, & p(0) &= N_A, \\ n(L) &= N_D, & p(L) &= \frac{n_i^2}{N_D}. \end{aligned}$$

Thus, the applied voltage enters through the electrostatic potential boundary conditions, while the carrier boundary values remain fixed by the contact doping concentrations.

4.3. Classical Gummel Iteration. The fully coupled drift–diffusion–Poisson system is nonlinear due to the coupling between the electrostatic potential, carrier densities, current densities, and recombination terms. To solve this system numerically, a classical Gummel iteration is employed.

At iteration k , the previous approximations are denoted by

$$\phi^{(k)}, \quad n^{(k)}, \quad p^{(k)}.$$

Rather than solving the entire nonlinear system simultaneously, the Gummel method [Gum64] decouples the problem into sequential subproblems for the electrostatic potential, electron density, and hole density. Each variable is updated while treating the remaining quantities as temporarily fixed.

4.3.1. Poisson Solver. Using the previous carrier densities $n^{(k)}$ and $p^{(k)}$, the Poisson equation

$$-\varepsilon\phi_{xx} = q(p - n + C)$$

is solved for the updated electrostatic potential $\phi^{(k+1)}$.

Using a second-order finite-difference discretization, the interior nodes satisfy

$$-\phi_{i-1}^{(k+1)} + 2\phi_i^{(k+1)} - \phi_{i+1}^{(k+1)} = \frac{qh^2}{\varepsilon} (p_i^{(k)} - n_i^{(k)} + C_i).$$

The boundary conditions are imposed by

$$\phi_0^{(k+1)} = \phi_0, \quad \phi_m^{(k+1)} = \phi_L.$$

This step updates the electrostatic field using the current approximation of the carrier densities and we obtained the following tridiagonal linear system.

$$\begin{bmatrix} 1 & 0 & 0 & \cdots & 0 & 0 \\ -1 & 2 & -1 & \cdots & 0 & 0 \\ 0 & -1 & 2 & \ddots & 0 & 0 \\ \vdots & \vdots & \ddots & \ddots & -1 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \phi_0^{(k+1)} \\ \phi_1^{(k+1)} \\ \vdots \\ \phi_{m-1}^{(k+1)} \\ \phi_m^{(k+1)} \end{bmatrix} = \frac{qh^2}{\varepsilon} \begin{bmatrix} \phi_0 \\ p_1^{(k)} - n_1^{(k)} + C_1 \\ \vdots \\ p_{m-1}^{(k)} - n_{m-1}^{(k)} + C_{m-1} \\ \phi_L \end{bmatrix}.$$

4.3.2. Recombination & Continuity Updates. After updating the electrostatic potential, the Shockley–Read–Hall recombination term is computed using the previous carrier densities:

$$R_i^{(k)} = \frac{n_i^{(k)} p_i^{(k)} - n_i^2}{\tau_p(n_i^{(k)} + n_1) + \tau_n(p_i^{(k)} + p_1)}.$$

In the present simulation, the midgap trap assumption

$$n_1 = n_i, \quad p_1 = n_i$$

is used. After updating the recombination term, the electron and hole continuity equations are solved sequentially using the Scharfetter–Gummel discretization. Using the updated potential $\phi^{(k+1)}$, the electron continuity equation

$$\frac{dJ_n}{dx} = qR$$

is solved for the updated electron density $n^{(k+1)}$.

The electron current density is discretized using the Scharfetter–Gummel flux:

$$J_{n,i+1/2} = \frac{qD_n}{h} \left[n_{i+1} B(\psi_{i+1/2}) - n_i B(-\psi_{i+1/2}) \right],$$

where

$$\psi_{i+1/2} = \frac{\phi_{i+1} - \phi_i}{V_T},$$

and

$$B(z) = \frac{z}{e^z - 1}$$

denotes the Bernoulli function.

The discrete continuity equation becomes

$$J_{n,i+1/2}^{(k+1)} - J_{n,i-1/2}^{(k+1)} = qhR_i^{(k)}.$$

The boundary conditions are

$$n_0^{(k+1)} = n_0, \quad n_m^{(k+1)} = n_L.$$

After substitution of the Scharfetter–Gummel fluxes, the following tridiagonal linear system is obtained for the updated electron density $n^{(k+1)}$.

$$\begin{bmatrix} 1 & 0 & 0 & \cdots & 0 & 0 \\ a_1 & b_1 & c_1 & \cdots & 0 & 0 \\ 0 & a_2 & b_2 & \ddots & 0 & 0 \\ \vdots & \vdots & \ddots & \ddots & c_{m-2} & 0 \\ 0 & 0 & 0 & a_{m-1} & b_{m-1} & c_{m-1} \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} n_0^{(k+1)} \\ n_1^{(k+1)} \\ \vdots \\ n_{m-1}^{(k+1)} \\ n_m^{(k+1)} \end{bmatrix} = \frac{h^2}{D_n} \begin{bmatrix} n_0 \\ R_1^{(k)} \\ \vdots \\ R_{m-1}^{(k)} \\ n_m \end{bmatrix}$$

where,

$$\begin{aligned} a_i &= B(-\psi_{i-1/2}), \\ b_i &= -[B(-\psi_{i+1/2}) + B(\psi_{i-1/2})], \\ c_i &= B(\psi_{i+1/2}). \end{aligned}$$

Next, using the updated electrostatic potential $\phi^{(k+1)}$, the hole continuity equation

$$\frac{dJ_p}{dx} = -qR$$

is solved for the updated hole density $p^{(k+1)}$. The hole continuity equation is discretized analogously using the same Scharfetter–Gummel flux approximation.

$$J_{p,i+1/2} = \frac{qD_p}{h} [p_{i+1}B(-\psi_{i+1/2}) - p_iB(\psi_{i+1/2})].$$

The discrete continuity equation becomes

$$J_{p,i+1/2}^{(k+1)} - J_{p,i-1/2}^{(k+1)} = -qhR_i^{(k)}.$$

The boundary conditions are

$$p_0^{(k+1)} = p_0, \quad p_m^{(k+1)} = p_L$$

This again produces the following tridiagonal linear system for the updated hole density $p^{(k+1)}$.

$$\begin{bmatrix} 1 & 0 & 0 & \cdots & 0 & 0 \\ \alpha_1 & \beta_1 & \gamma_1 & \cdots & 0 & 0 \\ 0 & \alpha_2 & \beta_2 & \ddots & 0 & 0 \\ \vdots & \vdots & \ddots & \ddots & \gamma_{m-2} & 0 \\ 0 & 0 & 0 & \alpha_{m-1} & \beta_{m-1} & \gamma_{m-1} \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} p_0^{(k+1)} \\ p_1^{(k+1)} \\ \vdots \\ p_{m-1}^{(k+1)} \\ p_m^{(k+1)} \end{bmatrix} = -\frac{h^2}{D_p} \begin{bmatrix} p_0 \\ R_1^{(k)} \\ \vdots \\ R_{m-1}^{(k)} \\ p_m \end{bmatrix}$$

where,

$$\begin{aligned}\alpha_i &= B(\psi_{i-1/2}), \\ \beta_i &= -\left[B(\psi_{i+1/2}) + B(-\psi_{i-1/2})\right], \\ \gamma_i &= B(-\psi_{i+1/2}).\end{aligned}$$

The hole continuity equation therefore also produces a tridiagonal linear system, with coefficients determined by the Scharfetter–Gummel discretization of the hole flux.

After each nonlinear iteration, convergence of the coupled system is monitored through the electrostatic potential update together with stabilization of the carrier densities and current profiles.

$$\begin{aligned}\text{err}_\phi^{(k)} &= \|\phi^{(k+1)} - \phi^{(k)}\|_\infty, \\ \text{err}_n^{(k)} &= \|n^{(k+1)} - n^{(k)}\|_\infty, \\ \text{err}_p^{(k)} &= \|p^{(k+1)} - p^{(k)}\|_\infty.\end{aligned}$$

Monitoring all three quantities separately is important because the electrostatic potential and carrier densities generally evolve on different numerical scales and may converge at different rates during the nonlinear iteration. The iterative process is repeated until the prescribed convergence criteria are satisfied.

At steady state, the total current

$$J_{\text{tot}} = J_n + J_p$$

should remain approximately constant throughout the device. Numerical current conservation was therefore monitored during the simulations as an additional diagnostic check. Numerically, this property is verified by monitoring the flatness of the total current profile across the computational domain. This serves as an additional consistency check for the nonlinear solver and the Scharfetter–Gummel discretization.

4.4. Numerical Difficulties and Stiffness Behavior. The classical Gummel iteration becomes increasingly difficult to solve as the doping magnitude or applied bias increases. In strongly stiff regimes, the nonlinear coupling between the Poisson and

continuity equations frequently produced ill-conditioned linear systems and numerical instability.

4.4.1. Increasing Doping Magnitude. Table 2 shows the convergence behavior under increasing symmetric doping concentrations with fixed applied bias $V_{\text{app}} = 0.01$.

Doping	Converged	Iterations	FinalError	Status
1e+15	1	12	3.56738527607092e-11	Converged
2e+15	1	103	9.1785579137138e-11	Converged
5e+15	0	11	NaN	Overflow
7e+15	0	11	NaN	Overflow
1e+16	0	5	NaN	Overflow

Table 2. Convergence behavior under increasing doping magnitude.

For doping levels 10^{15} and 2×10^{15} , the classical Gummel iteration converged successfully. However, beginning near 5×10^{15} , the iteration failed due to overflow behavior and loss of numerical stability. This reflects the increasing stiffness produced by stronger built-in electric fields and sharper carrier gradients inside the depletion region.

4.4.2. Increasing Applied Bias. Since the first experiment already showed instability near doping magnitudes of order 10^{16} , the applied-bias sweep in Table 3 was performed with the doping fixed at $N_A = N_D = 10^{15} \text{ cm}^{-3}$.

AppliedBias	Converged	Iterations	FinalError	Status
0	1	12	2.3103630120147e-11	Converged
0.01	1	12	3.56738527607092e-11	Converged
0.03	1	12	8.61241356009401e-11	Converged
0.05	1	13	4.45034287199775e-11	Converged
0.07	1	14	2.92994517536727e-11	Converged
0.1	1	15	4.66100213980525e-11	Converged
0.15	1	18	6.92847168526356e-11	Converged
0.2	1	24	7.89007886803361e-11	Converged
0.3	1	105	9.29517435244165e-11	Converged
0.4	0	17	NaN	Overflow

Table 3. Convergence behavior under increasing applied bias.

In this regime, the solver remained stable for small and moderate forward bias values, but the iteration count increased significantly as the applied voltage increased. Near $V_{\text{app}} = 0.4$, the iteration again failed due to overflow and loss of numerical stability. These observations motivate the quasi-linearized formulation derived in Chapter-2 of this thesis.

4.5. Quasi-Linearized Gummel Iteration. The numerical experiments of the previous section showed that the classical Gummel iteration becomes increasingly unstable in strongly stiff transport regimes. To improve the numerical stability of the nonlinear solver, the quasi-linearized formulation developed earlier in Section 3.3 was incorporated into the non-equilibrium drift-diffusion system. Rather than solving the nonlinear Poisson equation through the standard classical update, the quasi-linearized formulation partially incorporates the nonlinear carrier dependence directly into the Poisson solve itself. The resulting iteration therefore still follows the same overall Gummel framework, which are:

- (1) solve the Poisson equation,
- (2) update the recombination term,
- (3) solve the electron continuity equation,
- (4) solve the hole continuity equation,

but now the Poisson solve is replaced by the quasi-linearized formulation derived previously. And all remaining parts of the algorithm, including the Scharfetter-Gummel discretization, recombination update, continuity solves, and boundary conditions, remain unchanged from the previous classical Gummel formulation.

4.5.1. Equilibrium-Based Initialization of Gummel Iteration. The non-equilibrium drift-diffusion system was initialized using the converged equilibrium solution obtained from the quasi-linearized equilibrium solver developed in Chapter 3.

In particular, the equilibrium profiles

$$\phi_{\text{eq}}, \quad n_{\text{eq}}, \quad p_{\text{eq}}$$

were used as the initial guess for the non-equilibrium iteration.

This approach is physically motivated since low-bias non-equilibrium states remain close to thermodynamic equilibrium. Numerically, the equilibrium solution provides a substantially more stable initialization than naive global linear guesses or constant carrier profiles.

In the MATLAB implementation, the converged equilibrium solution was imported directly into the non-equilibrium solver and used as the starting iterate for the quasi-linearized Gummel iteration. The full non-equilibrium system, including applied bias and Shockley–Read–Hall recombination, was then solved using this equilibrium–based initialization.

4.6. Forward–Bias Behavior and Simulation. In this section, the coupled drift–diffusion–Poisson system is investigated under forward-bias conditions using the quasi-linearized Gummel iteration together with the Scharfetter–Gummel flux discretization. Unless otherwise stated, the simulations presented in this section were performed using an applied forward bias of

$$V_{\text{app}} = 0.20 \text{ V},$$

with nonlinear convergence tolerances chosen as

$$\varepsilon_{\phi} = 10^{-10}, \quad \varepsilon_n = \varepsilon_p = 10^4.$$

The equilibrium solution obtained previously was used as the initial condition for the nonlinear iteration process. In addition, Shockley–Read–Hall (SRH) recombination was included using the recombination lifetimes

$$\tau_n = \tau_p = 5 \times 10^{-9} \text{ s}.$$

Under forward bias, the depletion barrier decreases and carrier injection across the junction interface becomes significantly enhanced. Consequently, strong carrier

transport and recombination activity develop near the depletion region, producing clear non-equilibrium behavior throughout the coupled drift–diffusion system.

4.6.1. Nonlinear Convergence Behavior. The convergence behavior of the quasi-linearized Gummel iteration under forward bias is shown in Figure 6.

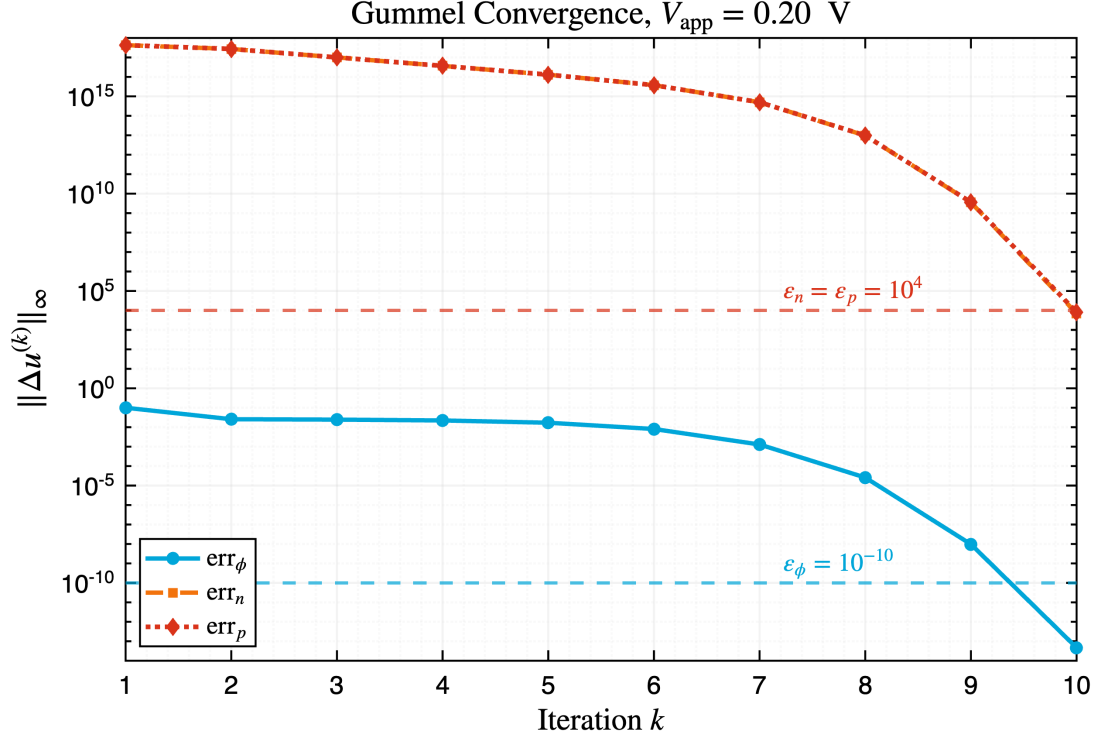


Figure 6. Convergence history of the quasi-linearized Gummel iteration under forward bias.

The convergence history demonstrates stable nonlinear behavior throughout the iteration process. The electrostatic potential error decreases rapidly as the iteration approaches the non-equilibrium steady-state solution, while the electron and hole density errors evolve more gradually due to the coupled drift–diffusion transport equations.

The electron and hole error curves exhibit strong overlap throughout most of the iteration process. This behavior reflects both the symmetric doping structure of the

abrupt p–n junction and the strong nonlinear coupling between the carrier transport equations.

The convergence history exhibits two distinct phases. During the first several iterations, all three error measures remain relatively unchanged and form an initial plateau region. Beyond approximately the seventh iteration, the errors begin to decrease rapidly as the nonlinear iteration approaches the non-equilibrium steady-state solution.

The electrostatic potential error decreases more rapidly than the carrier density errors and reaches the prescribed tolerance within approximately ten iterations. In contrast, the electron and hole density errors evolve more gradually because they remain strongly coupled through the drift–diffusion transport equations.

The use of equilibrium-based initialization significantly improves the convergence behavior of the non-equilibrium solver. Since the converged equilibrium solution already captures the built-in electrostatic structure of the junction, only a relatively small number of Gummel iterations are required after the external bias is introduced.

Overall, the results demonstrate that the quasi-linearized Gummel iteration provides a stable and computationally efficient approach for computing non-equilibrium steady-state solutions of the coupled drift–diffusion–Poisson system under forward-bias operation.

4.6.2. SRH Recombination Dynamics. Under non-equilibrium forward-bias conditions, electron and hole populations simultaneously coexist near the junction interface, producing carrier recombination within the depletion region. In the present work, recombination is modeled using the Shockley–Read–Hall (SRH) recombination relation

$$R = \frac{np - n_i^2}{\tau_p(n + n_1) + \tau_n(p + p_1)},$$

where n and p denote the electron and hole carrier concentrations, n_i is the intrinsic carrier concentration, and τ_n and τ_p represent the electron and hole recombination lifetimes, respectively. The quantities n_1 and p_1 correspond to the SRH trap reference densities under the midgap trap approximation.

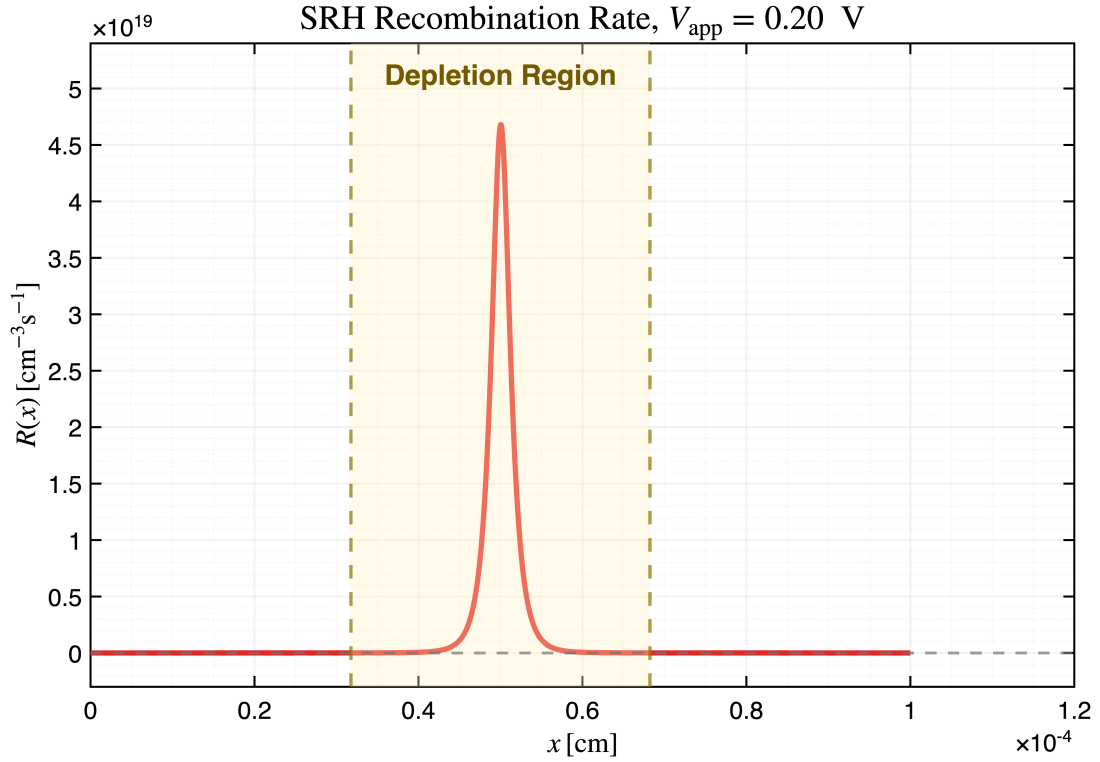


Figure 7. Shockley–Read–Hall recombination profile under forward bias.

The resulting recombination profile is strongly localized near the depletion region, where the overlap between injected electron and hole populations becomes largest under forward bias. In particular, the quantity

$$np - n_i^2$$

increases significantly near the junction interface, producing the sharply concentrated recombination peak observed in Figure 7. The peak occurs near the center of the depletion region and is confined to a relatively narrow spatial interval, indicating that

recombination activity is highly concentrated rather than uniformly distributed throughout the device.

Away from the depletion region, one carrier species dominates while the other becomes comparatively small. Consequently, the product np decreases substantially and the recombination rate rapidly approaches zero within the quasi-neutral regions of the device.

The localized recombination peak is physically consistent with forward-biased p–n junction transport. Under forward bias, electrons diffuse into the p-side while holes diffuse into the n-side, increasing the overlap between the two carrier populations near the junction interface and producing enhanced recombination activity within the depletion region.

4.6.3. Current Conservation and Flux Transport. The electron, hole, and total current densities obtained from the Scharfetter–Gummel discretization are shown in Figure 8. Under steady-state transport conditions, the drift–diffusion continuity equations satisfy

$$\frac{\partial J_n}{\partial x} = qR, \quad \frac{\partial J_p}{\partial x} = -qR,$$

where R denotes the Shockley–Read–Hall recombination rate. Adding the two equations yields

$$\frac{\partial}{\partial x}(J_n + J_p) = 0,$$

which implies that the total current density should remain spatially conserved throughout the device.

Consequently, although the individual electron and hole current components may vary significantly throughout the device, their sum should remain approximately constant. Figure 8 illustrates this behavior.

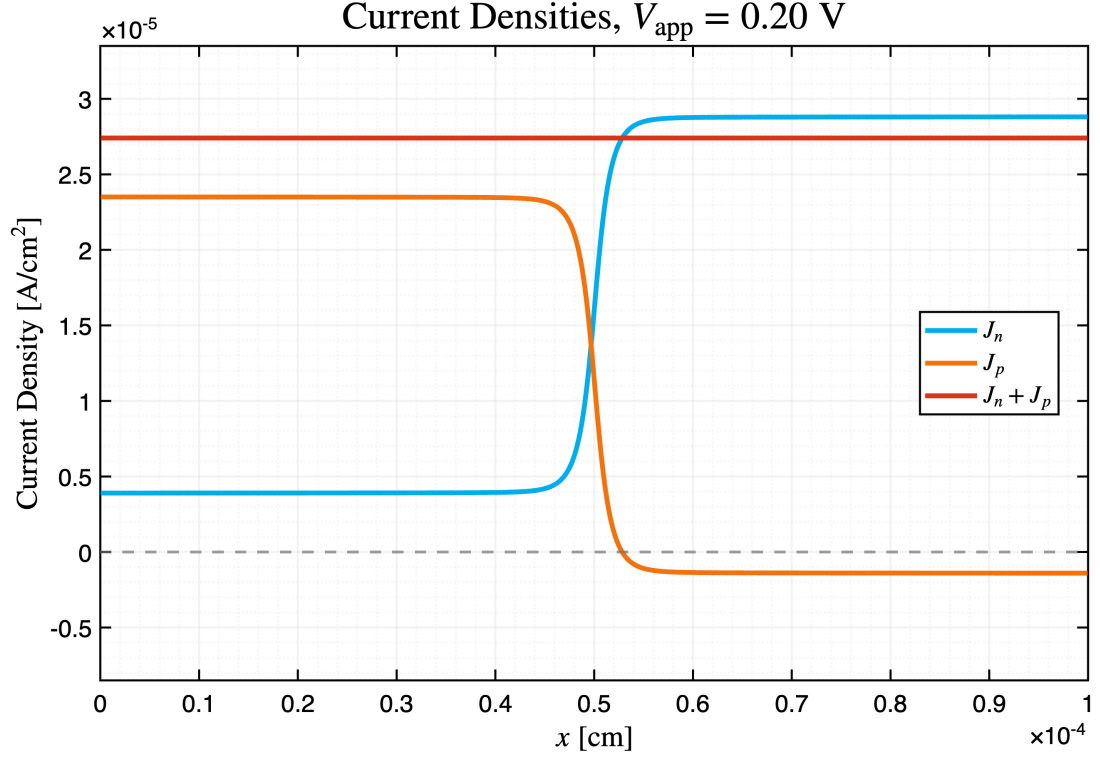


Figure 8. Electron, hole, and total current densities under forward-bias conditions.

This current conservation behavior is clearly observed in Figure 8, where the total current density remains almost perfectly flat across the computational domain. Despite substantial spatial variation in the individual fluxes, their combined contribution remains invariant. Near the depletion region, the electron current density increases while the hole current density decreases, indicating a gradual transfer of current dominance between holes and electrons across the junction.

Consequently, this behavior of the total current density provides an important numerical consistency verification for the coupled drift–diffusion–Poisson solver and demonstrates that the discretized SG flux formulation preserves current conservation with good numerical accuracy.

4.7. Reverse-Bias Behavior and Simulation. In this section, the coupled drift–diffusion–Poisson system is investigated under reverse-bias conditions using the quasi-linearized Gummel iteration together with the Scharfetter–Gummel flux discretization. Unless otherwise stated, the simulations presented in this section were performed using an applied reverse bias of

$$V_{\text{app}} = -0.20 \text{ V},$$

with nonlinear convergence tolerances chosen as

$$\varepsilon_\phi = 10^{-10}, \quad \varepsilon_n = \varepsilon_p = 10^4.$$

The equilibrium solution obtained previously was used as the initial condition for the nonlinear iteration process. In addition, Shockley–Read–Hall (SRH) recombination was included using the recombination lifetimes

$$\tau_n = \tau_p = 5 \times 10^{-9} \text{ s}.$$

Under reverse bias, the depletion region widens and carrier injection across the junction becomes increasingly suppressed. Consequently, the numerical behavior differs noticeably from the forward-bias simulations, particularly in the carrier transport and recombination dynamics near the junction interface.

4.7.1. Nonlinear Convergence Behavior. The convergence history of the reverse-bias simulation is shown in Figure 9. The convergence history exhibits an initial transient phase followed by a rapid error-reduction regime. During the first several iterations, the carrier-density errors exhibit a mild overshoot and then remain relatively unchanged over a broad range of iterations before eventually decreasing rapidly toward the prescribed tolerances. The electrostatic potential error exhibits a similar plateau behavior but converges to its tolerance more quickly than the carrier-density errors.

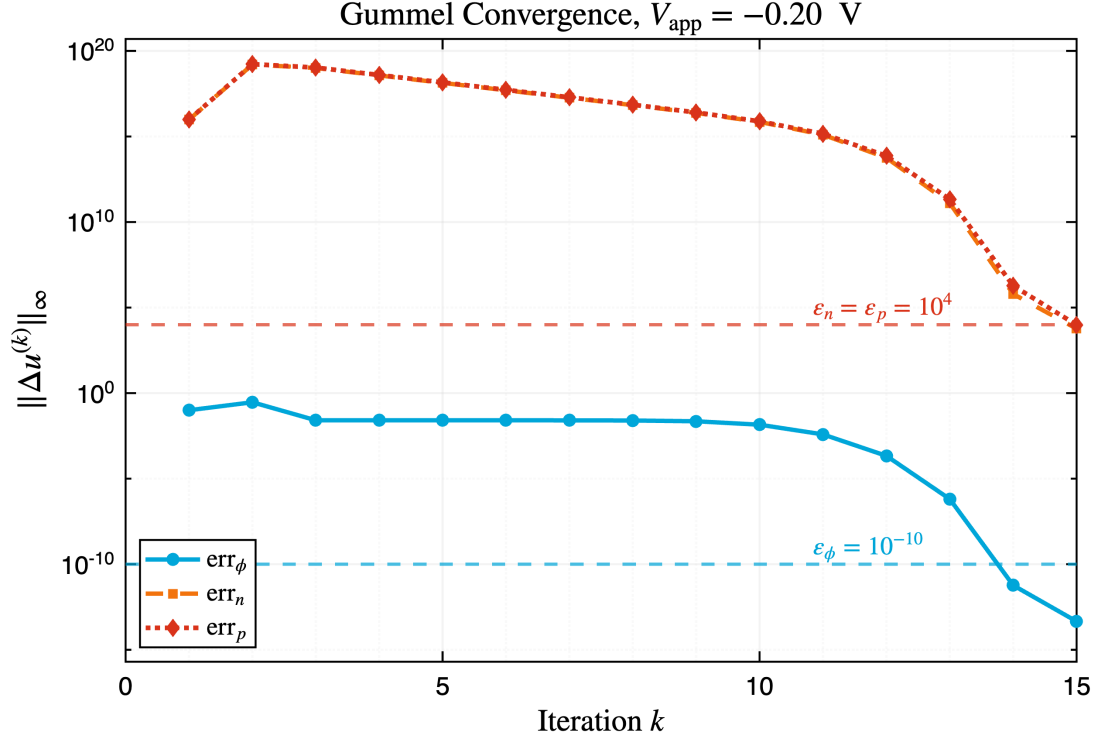


Figure 9. Convergence history of the quasi-linearized Gummel iteration under reverse bias.

The initial overshoot is associated with the stronger carrier redistribution induced by reverse bias. As the depletion region expands, minority carriers near the junction interface become increasingly suppressed, requiring a larger nonlinear adjustment during the early stages of the iteration process. Once the iterates move sufficiently close to the reverse-biased steady-state solution, all error components decrease rapidly.

The electron and hole error curves remain strongly overlapped throughout most of the iteration process. This behavior is consistent with the symmetric structure of the abrupt p-n junction and reflects the strong nonlinear coupling between the carrier transport equations.

Despite the temporary overshoot and prolonged plateau phase, the nonlinear iteration remains stable throughout the simulation and converges successfully after

several updates. The results demonstrate that the equilibrium-based initialization continues to provide an effective starting point even under reverse-bias operation and that the quasi-linearized Gummel iteration maintains robust convergence for the coupled nonlinear drift–diffusion–Poisson system.

4.7.2. SRH Generation Dynamics. The reverse-bias SRH recombination profile is shown in Figure 10.

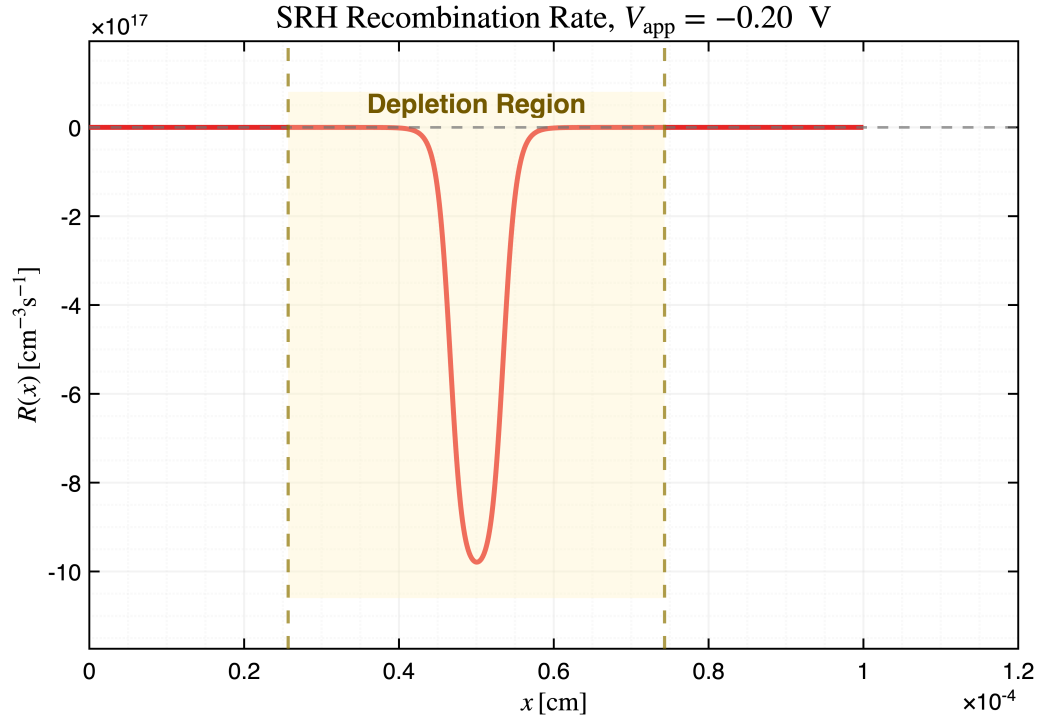


Figure 10. SRH generation profile under reverse-bias conditions.

The Shockley–Read–Hall recombination model is given by

$$R = \frac{np - n_i^2}{\tau_p(n + n_1) + \tau_n(p + p_1)},$$

where n and p denote the electron and hole carrier densities, respectively, and τ_n, τ_p represent the carrier recombination lifetimes.

Under reverse bias, the overlap between electrons and holes gets separated near the interface due to the widening of the depletion region. Consequently, the quantity

$$np - n_i^2$$

becomes negative within the depletion region, producing a negative SRH profile corresponding to net carrier generation rather than recombination.

The resulting generation profile remains strongly localized within the depletion region, where the carrier concentrations become sufficiently depleted that

$$np < n_i^2.$$

Consequently, the SRH term becomes negative and a localized generation region forms near the junction interface. Away from the junction interface, the carrier populations become increasingly separated and the SRH contribution rapidly approaches zero within the quasi-neutral regions.

The localized negative SRH profile observed in the simulation is physically consistent with reverse-biased p–n junction behavior. Under reverse bias, the depletion region widens and the electron and hole populations become increasingly separated. As a result, the product np falls below its equilibrium value n_i^2 , suppressing recombination and producing net carrier generation within the depletion region. The generated carriers are subsequently swept apart by the built-in electric field, contributing to the reverse-bias current.

4.7.3. Current Conservation and Flux Transport. The electron, hole, and total current densities obtained under reverse bias are shown in Figure 11.

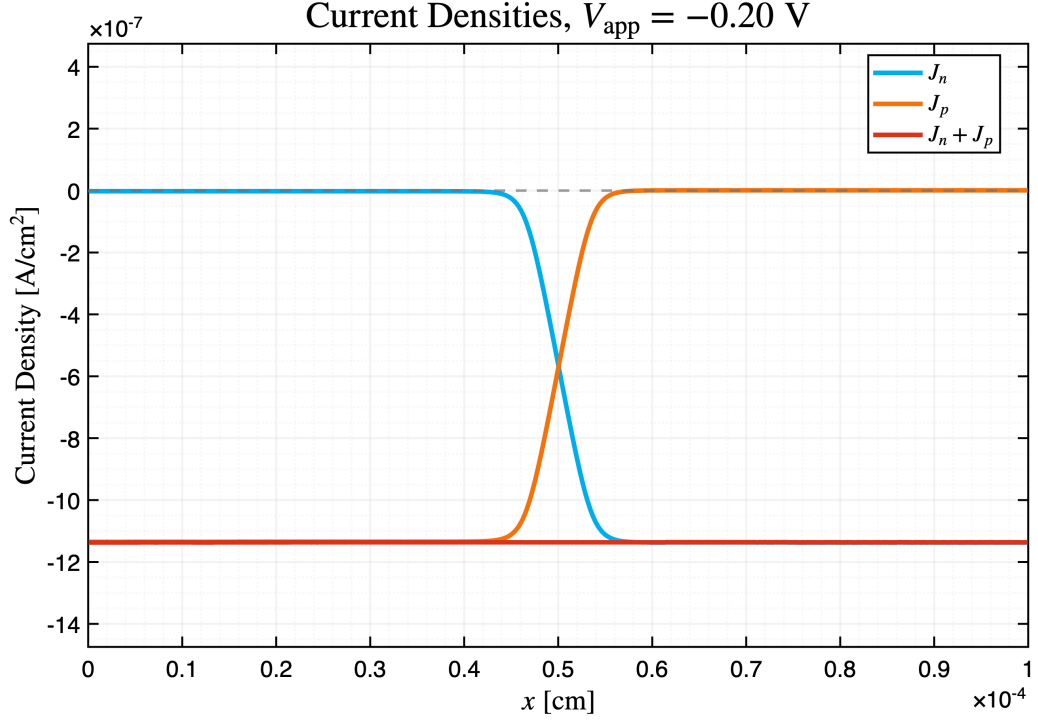


Figure 11. Electron, hole, and total current densities under reverse-bias conditions.

The total current density is computed from

$$J_{\text{tot}} = J_n + J_p,$$

where J_n and J_p denote the electron and hole current densities, respectively.

Under reverse bias, the electron and hole current components separate strongly across the junction interface. The electron current becomes dominant on the n-side of the device, while the hole current dominates on the p-side. The transition between the two current components occurs primarily in the vicinity of the junction interface and reflects the widening of the depletion region and the suppression of carrier injection under reverse bias.

Despite the substantial spatial variation in the individual carrier currents, their sum remains nearly constant throughout the computational domain. In particular, the total

current density remains approximately equal to a constant negative value, consistent with the steady-state continuity relation

$$\frac{\partial J_{\text{tot}}}{\partial x} = 0.$$

This invariant behavior of J_{tot} provides an important numerical consistency check for the coupled drift–diffusion solver. It demonstrates that the Scharfetter–Gummel discretization accurately captures the transfer of current between electron and hole transport channels while preserving current conservation with good numerical accuracy under reverse-bias operation.

4.8. Current–Voltage Characteristics of a PN Junction Diode. After separately studying the forward-bias and reverse-bias simulations, the final numerical experiment was to compute the overall current–voltage behavior of the p–n junction diode. For each applied voltage V_{app} , the coupled drift–diffusion–Poisson system was solved using the Scharfetter–Gummel discretization together with the quasi-linearized Gummel iteration. After convergence, the electron and hole current densities were evaluated on the cell edges, and the total current density was obtained from

$$J_{\text{tot}} = J_n + J_p.$$

The quantity plotted in this section is the current density rather than the total terminal current. Consequently, the simulation produces a J – V characteristic instead of a traditional I – V characteristic. The total current is related to the current density through

$$I = J_{\text{tot}}A,$$

where A denotes the cross-sectional area of the device. Since the present model is one-dimensional and does not assume a specific device geometry, plotting the current density is the more natural choice.

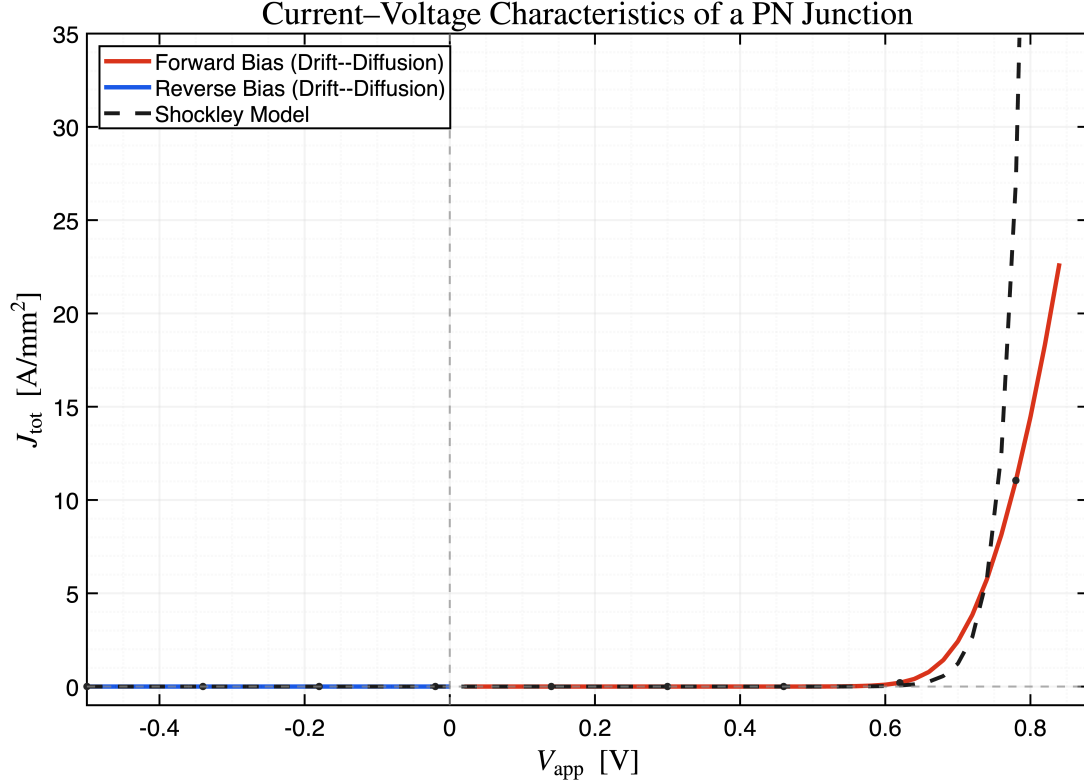


Figure 12. Current–voltage characteristics of the p–n junction diode

4.8.1. Forward-Bias Behavior. Under forward bias, the numerical solution exhibits the expected diode turn-on behavior. As the applied voltage increases, the depletion region narrows and carrier injection across the junction increases significantly. Consequently, the total current density rises rapidly once the applied voltage approaches the typical turn-on region of a silicon p–n junction diode.

Figure 12 shows that the numerical curve begins to increase sharply near approximately 0.6–0.7 V, which is consistent with the standard knee-voltage behavior of silicon diodes. This demonstrates that the numerical drift–diffusion solver successfully reproduces the expected rectifying behavior of the device.

The forward-bias branch also displays the characteristic nonlinear growth predicted by the Shockley diode relation, which is discussed in a section below, after Reverse-Bias Behavior .

4.8.2. Reverse-Bias Behavior. Under reverse bias, the numerical current density remains small in magnitude and approximately saturated. Physically, reverse bias widens the depletion region and suppresses carrier injection across the junction, leaving only a small reverse leakage current.

The reverse-bias branch therefore remains relatively flat compared to the forward-bias regime. This behavior is consistent with the expected operation of a p–n junction prior to breakdown.

The present model does not include avalanche breakdown or Zener tunneling effects [06]. Consequently, the reverse-bias current remains in the saturation regime and does not exhibit breakdown at large negative voltages. Modeling breakdown behavior would require additional mechanisms that are beyond the scope of the present drift–diffusion–Poisson formulation.

4.8.3. Comparison with the Shockley Diode Law. To compare the numerical drift–diffusion solution with the classical diode theory, the Shockley diode relation was also included [06].

$$J_{\text{Shockley}} = J_s \left(e^{V_{\text{app}}/V_T} - 1 \right),$$

where $V_T = k_B T/q$ is the thermal voltage and J_s is the reverse saturation current density. To ensure a meaningful comparison, J_s was computed using the same physical parameters employed throughout the numerical simulation. Using the diffusion lengths

$$L_n = \sqrt{D_n \tau_n}, \quad L_p = \sqrt{D_p \tau_p},$$

the theoretical saturation current density is approximated by

$$J_s = q n_i^2 \left(\frac{D_n}{L_n N_A} + \frac{D_p}{L_p N_D} \right),$$

where D_n and D_p denote the electron and hole diffusion coefficients, τ_n and τ_p are the carrier recombination lifetimes, N_A and N_D are the acceptor and donor doping concentrations, and n_i is the intrinsic carrier concentration. In this way, both the analytical Shockley model and the numerical drift–diffusion model are compared using consistent physical parameters. The comparison with the Shockley diode relation serves as a useful validation of the numerical model.

As shown in Figure 12, both curves display the same qualitative diode behavior, including reverse-current saturation and exponential current growth under forward bias. In particular, both models exhibit a clear turn-on region near approximately 0.6–0.7 V, which is characteristic of silicon p–n junction diodes.

Some quantitative differences between the two curves are expected. As the applied voltage becomes large, the numerical drift–diffusion solution predicts a somewhat smaller current density than the ideal Shockley model. This discrepancy arises because the Shockley relation is an idealized analytical model derived under some assumptions [06], whereas the present numerical solver resolves the full coupled electrostatic and carrier transport behavior throughout the device. In particular, the drift–diffusion model accounts for the spatially varying carrier densities, electrostatic potential distributions, and Shockley–Read–Hall recombination processes across the junction. Overall, the close qualitative agreement between the numerical and analytical curves indicates that the Scharfetter–Gummel drift–diffusion solver successfully reproduces the essential current–voltage characteristics of a silicon p–n junction diode while simultaneously resolving the detailed internal transport physics.

5. Conclusion

The drift–diffusion–Poisson system provides a rich mathematical framework for studying nonlinear semiconductor transport phenomena. Throughout this thesis, equilibrium and non-equilibrium p–n junction models were investigated through a combination of mathematical analysis, numerical discretization, and computational simulation.

One of the central difficulties of the system is the strong nonlinear coupling between electrostatic potential, carrier transport, and recombination behavior. As the applied bias and doping magnitude increase, the transport equations become increasingly stiff and difficult to solve numerically. Much of the present work therefore focused on constructing numerical schemes capable of maintaining stability and physically meaningful transport behavior in these strongly nonlinear regimes.

The simulations demonstrate that quasi-linearized drift–diffusion formulations combined with Scharfetter–Gummel discretization provide a robust framework for semiconductor transport simulation. Within this framework, numerical methods were developed and implemented for both equilibrium and non-equilibrium p–n junction problems. The resulting simulations successfully reproduce several important qualitative features of semiconductor devices, including depletion-region formation, carrier recombination and generation behavior, forward- and reverse-bias transport, and the characteristic nonlinear current–voltage response of a silicon p–n junction diode. The numerical current–voltage characteristics also exhibit good qualitative agreement with the classical Shockley diode relation, providing additional validation of the computational framework.

At the same time, the present work also highlights several limitations and possible extensions of the model. The current formulation does not include optical carrier generation, high-field breakdown effects, or more advanced mobility models. Extending

the framework to include external light sources would allow the numerical investigation of photovoltaic and solar-cell devices, where additional carrier generation mechanisms strongly influence transport behavior inside the semiconductor. Such extensions would naturally produce more complicated nonlinear systems and introduce additional numerical challenges.

Overall, the framework developed in this thesis provides a foundation for studying a broader class of nonlinear semiconductor transport problems arising in computational electronics and applied partial differential equations. More generally, the present work illustrates how mathematical analysis, discretization, and numerical simulation can be combined together to investigate complex systems of strongly coupled nonlinear partial differential equations arising in physics or other areas of the applied sciences.

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